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1 Why we estimate this equation: the HSA Phillips curve

Our estimating equation is the theory-motivated counterpart of the New Keynesian Phillips curve derived by Fujiwara–Matsuyama (2026, *Journal of Monetary Economics*) under the HSA demand system. This section states their result, maps it to what we take to data, and makes explicit how far the empirical specification departs from the theory.

1.1 The HSA Phillips curve

A final-good producer aggregates a continuum of varieties through an HSA demand system with market-share function $s(z)$, where $z = p/A$ is a firm’s price normalized by the single price aggregator A (the reference price faced by firms, an inverse index of competitive pressure). Intermediate firms set prices under a Rotemberg adjustment cost χ and enter endogenously with a one-period time-to-build lag. Solving the firm’s problem, PPI inflation follows (their Eq. 21)

$$\hat{\pi}_t = \beta(1 - \delta) \mathbb{E}_t \hat{\pi}_{t+1} + \underbrace{\frac{\zeta(z) - 1}{\chi}}_{\text{slope on marginal cost}} (\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t) - \underbrace{\frac{1}{\chi} \frac{1 - \rho(z)}{\rho(z)}}_{\text{entry / cost-push}} \hat{N}_t, \quad (1)$$

where $\hat{N}_t$ is the inverse HHI (number of firms), $\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t$ is the inverse markup (the model’s measure of real marginal cost), $\zeta(z) = 1 - s'(z)z/s(z) > 1$ is the price-elasticity function (so the flexible-price markup is $\mu = \zeta/(\zeta - 1)$), and $\rho(z)$ is the pass-through rate.

1.2 Two ways competition enters

(1) The slope is a function of the firm count (structural flattening). The coefficient on marginal cost, $(\zeta(z) - 1)/\chi$, depends on the price elasticity $\zeta(z)$, which through $z = p/A$ and the symmetric-equilibrium condition $s(z) = 1/N$ is a function of the firm count N . Rising concentration—*fewer firms*, higher markups, lower pass-through $\rho(z)$ —lowers this slope, so the Phillips curve flattens *structurally*. Linearizing the slope in the trend level N , $\bar{N}_t$ of the firm count gives

$$\kappa_t = \kappa_0 + \delta \bar{N}_t, \quad (2)$$

so a *positive* δ is the theory’s flattening prediction: the slope moves with the firm count and falls as concentration rises and N declines.

(2) Endogenous entry is a cost-push term (estimation bias). The second term $-\frac{1}{\chi} \frac{1 - \rho(z)}{\rho(z)} \hat{N}_t$ appears only away from CES. It means that the cyclical component of firm entry acts as a cost-push shock through strategic complementarity. Omitting it biases an empirical Phillips-curve estimate (omitted-variable problem). We retain the cyclical term explicitly with its own coefficient $\theta_t = \theta_0 + \gamma \bar{N}_t$.

CES is the irrelevant case. Under CES, $s(z) = \gamma_{\text{CES}} z^{1-\theta}$, so $\zeta(z) = \theta$ is constant and pass-through is $\rho(z) = 1$. The entry term vanishes and the slope is the constant $(\theta - 1)/\chi$: market concentration neither flattens the curve nor moves inflation. This is exactly our CES benchmark, and departing from CES (the HSA models) is what can make δ and the $\hat{N}_t$ term nonzero.

1.3 From theory to the estimating equation

Writing the activity/inverse-markup gap as x_t , adding a backward-looking inertia term (α) and a reduced-form shock e_t , and decomposing the observed firm count into trend and cycle,

$$N_t^{\text{obs}} = \bar{N}_t + \hat{N}_t + \nu_t, \quad (3)$$

where $\bar{N}_t$ is a random-walk-with-drift trend, $\hat{N}_t$ is a stationary AR(2) gap (truncated to the stationary region), and $\nu_t \sim \mathcal{N}(0, \sigma_N^2)$ is measurement error, the HSA Phillips curve becomes the equation we estimate:

$$\boxed{\pi_t = \alpha \pi_{t-1} + (1 - \alpha) \mathbb{E}_t \pi_{t+1} + \kappa_t x_t - \theta_t \hat{N}_t + e_t, \quad \kappa_t = \kappa_0 + \delta \bar{N}_t, \quad \theta_t = \theta_0 + \gamma \bar{N}_t.} \quad (4)$$

The trend $\bar{N}_t$ carries the structural-flattening channel (through δ); the cycle $\hat{N}_t$ carries the entry cost-push channel (through θ_t).

1.4 Distance from the theory

We call this exercise *semi-structural* rather than structural because the specification departs from (1) in several ways. To avoid over-reading the coefficients, we state each:

- the theory is about *PPI* inflation, but there is no PPI expectation series, so we pair observed PPI with a common external expectation series and the expectation term is not a matched structural object;
- the theoretical regressor is real marginal cost / the inverse markup, but the main results use the negative unemployment gap.
- we impose no cross-coefficient restrictions from the theory and approximate κ_t as a *linear* function of $\bar{N}_t$;
- we add backward-looking inertia (α), which is absent from (1);
- expected inflation is an external survey measure (Section 2);

2 Data

Sample. Quarterly, 1982Q1–2012Q4 ($T = 124$) in the US.

Inflation. We compare headline CPI, core CPI, and PPI. Expected inflation $\mathbb{E}\pi$ is the same external survey series in all specifications. Because the Survey of Professional Forecasters does not publish a PPI forecast, the PPI specifications use the same expectation series as CPI. The PPI estimates are therefore not a structural estimate matched through the expectation term, but a semi-structural external-validity check that changes only the observed inflation index. A measurement mismatch with the expectation series also remains for core CPI.

Activity gaps (five). Negative unemployment gap, BN output gap, HP output gap ($100 \times \log$ real output HP-filtered, in the same 100-log-point units as BN), HP labor-share gap, and inverse markup. The unemployment measure enters as the *negative* unemployment gap $x_t = -(u_t - u_t^*)$ —labor-market tightness, positive in expansions and negative in downturns (e.g. $x_t \approx -4$ in 2009), co-moving positively with the output gaps. A positive slope κ on x_t is therefore a conventional downward-sloping Phillips curve.

Competition proxy. The competition series is Gustavo’s firm-count measure (annual), i.e. the Inverse of the *HHI of U.S. listed firms*. The main results use PCHIP interpolation to a quarterly value observed every quarter; the additional results observe the annual value only in Q4 and infer the Q1–Q3 quarterly states inside the state-space model. The latter is a mixed-frequency specification that does not treat the interpolated value as data and propagates interpolation uncertainty into the posterior. *declines*.

Unit convention. The default transform is $(100 \log N - \overline{100 \log N})/10$, so one unit of $\bar{N}$ or $\hat{N}$ is a ten-log-point change around the sample mean. The reported $\delta, \theta, \theta_0, \gamma$ are all on this ten-log-point scale.

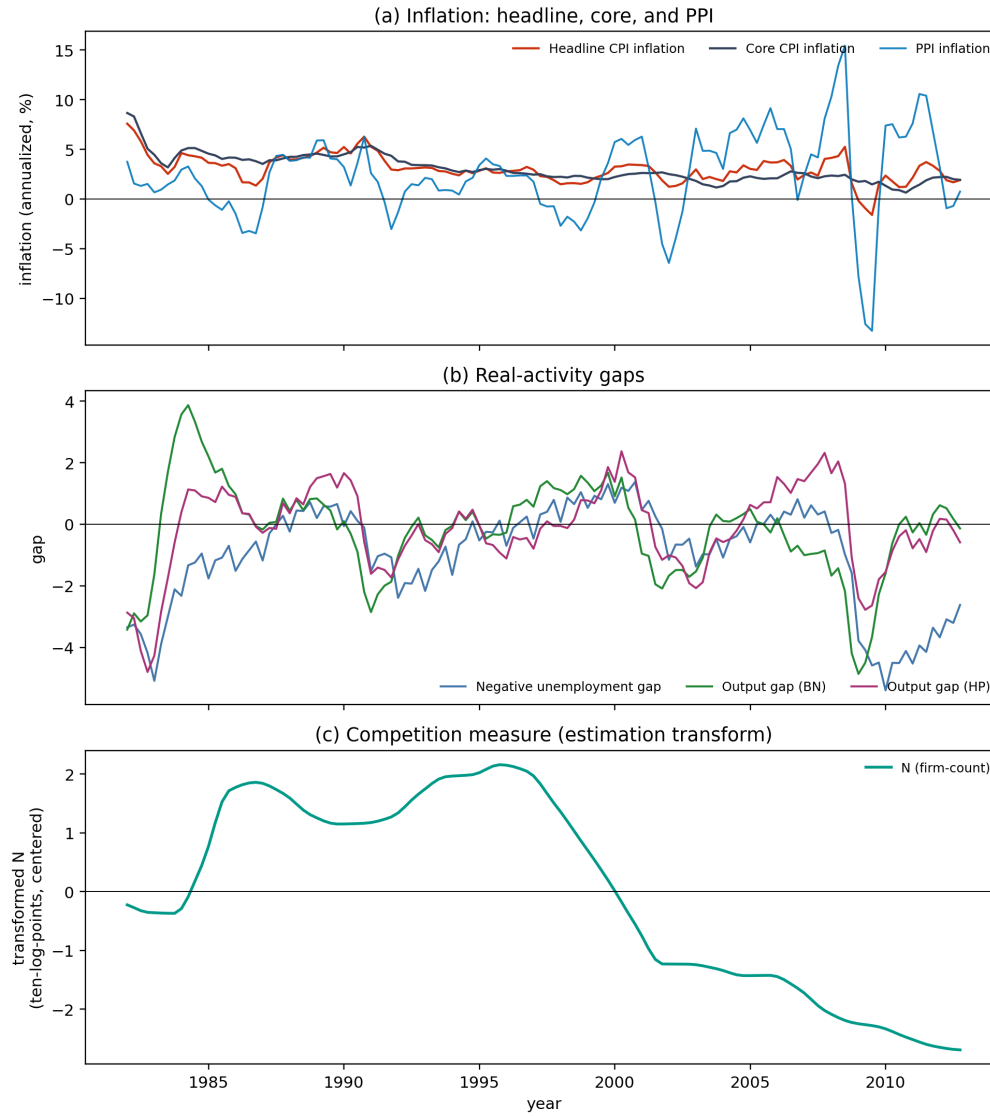


Figure 1: Data. (a) Headline, core, and PPI inflation. (b) Real-activity gaps (negative unemployment gap, BN and HP output gaps). (c) Competition measure (estimation transform of the listed-firm count).

3 Econometric model and estimation

3.1 Model variants

The five models differ only in the restriction on the slope (Table 1).

Table 1: Model variants.

Model	κ_t	θ_t	Role	Estimation
CES	constant	0	Textbook NKPC benchmark	Gibbs Sampling
HSA steady	$\kappa_0 + \delta \bar{N}_t$	0	$\bar{N}$ channel only	Gibbs Sampling + FFBS
HSA dynamic	constant	constant	$\hat{N}$ channel only	Gibbs Sampling + FFBS
HSA const-theta	$\kappa_0 + \delta \bar{N}_t$	constant	HSA full with $\gamma = 0$	Gibbs Sampling + FFBS
HSA full	$\kappa_0 + \delta \bar{N}_t$	$\theta_0 + \gamma \bar{N}_t$	HSA Full	Gibbs Sampling + Particle Filter

Estimator for HSA full. In this edition, **HSA full is estimated by a Particle Gibbs sampler** rather than the alternating FFBS used for the other models. Because the bilinear term $-\gamma \bar{N}_t \hat{N}_t$ makes the joint firm-count state non-linear-Gaussian. A single conditional sequential-Monte-Carlo (particle) updates the whole state path (512 particles); every coefficient, variance, and other block—and every other model—is unchanged, as are the number of chains, iterations, burn-in, and thinning.

3.2 The inflation equation actually estimated

To put the theoretical equation (4) directly in regression form, we move expected inflation to the left-hand side and define

$$y_t \equiv \pi_t - E_t \pi_{t+1}, \quad a_t \equiv \pi_{t-1} - E_t \pi_{t+1} \quad (5)$$

Making the unforecastable component of the activity variable explicit, the estimating system is

$$y_t = \alpha a_t + (\kappa_0 + \delta \bar{N}_t) x_t - (\theta_0 + \gamma \bar{N}_t) \hat{N}_t + \lambda_{e\zeta} \zeta_t + \eta_t, \quad (6)$$

$$x_t = \phi_1 x_{t-1} + \zeta_t, \quad \eta_t \perp \zeta_t. \quad (7)$$

That is, the original NKPC shock decomposes as $e_t = \lambda_{e\zeta} \zeta_t + \eta_t$. Estimating $\lambda_{e\zeta}$ allows a correlation between the contemporaneous activity shock and the inflation shock, rather than forcing a simple regression that treats x_t as exogenous. The saved correlation coefficient is

$$\rho_{e\zeta} = \frac{\lambda_{e\zeta} \sigma_\zeta}{\sqrt{\lambda_{e\zeta}^2 \sigma_\zeta^2 + \sigma_\eta^2}} \quad (8)$$

The main HSA steady sets $\theta_0 = \gamma = 0$; HSA dynamic sets $\delta = \gamma = 0$.

For numerical stability the implementation updates internal coefficients on the rescaled regressors $x_t/100$ and $x_t\bar{N}_t/100$ for κ_0 and δ ; the coefficients are rescaled back by 100 on saving, so the $\kappa_0, \delta, \kappa_t$ in the text, tables, and figures are all in the physical units of (6). They must not be multiplied by 100 or divided by 100 again.

3.3 State-space representation splitting the firm count into trend and cycle

The observation and transition equations for the firm count are

$$N_t^{\text{obs}} = \bar{N}_t + \hat{N}_t + \nu_t, \quad \nu_t \sim N(0, \sigma_N^2), \quad (9)$$

$$\hat{N}_t = \rho_1 \hat{N}_{t-1} + \rho_2 \hat{N}_{t-2} + u_t, \quad u_t \sim N(0, \sigma_u^2), \quad (10)$$

$$\bar{N}_t = n + \bar{N}_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2) \quad (11)$$

where $\bar{N}_t$ is the low-frequency trend with drift n and $\hat{N}_t$ is a stationary AR(2) cycle. Rather than a two-step procedure that first filters the observed firm count into components and then substitutes them into the NKPC, we extract both states using the firm-count equation and the inflation equation within the same Gibbs iteration.

Two observation frequencies. The PCHIP version uses (9) at all 124 quarters. The annual-Q4 version places the annual firm count in each year's Q4,

$$N_t^{\text{obs}} = \begin{cases} (100 \log N_{y(t)}^{\text{annual}} - \bar{c})/10, & t \in \mathcal{T}_N \text{ (Q4)}, \\ \text{missing}, & t \notin \mathcal{T}_N \text{ (Q1 -- Q3)}, \end{cases} \quad |\mathcal{T}_N| = 31, \quad (12)$$

where $\bar{c}$ is the same centering constant as the PCHIP reference series, keeping the coefficient units comparable. In Q4 the Kalman update uses two observation rows—the firm-count equation and the inflation equation—and in Q1–Q3 only the one inflation row. Because the state transition and prediction continue at every quarter, the Q1–Q3 $\bar{N}_t, \hat{N}_t$ are not simple interpolations but FFBS draws conditioned on the annual anchors, the state transition, and the inflation equation, updated every iteration. The inverse-gamma update of σ_N^2 likewise uses only the 31 finite Q4 residuals.

For the main HSA steady the state is

$$s_t = (\hat{N}_t, \hat{N}_{t-1}, \bar{N}_t)', \quad s_t = c + F s_{t-1} + w_t \quad (13)$$

with

$$c = \begin{bmatrix} 0 \\ 0 \\ n \end{bmatrix}, \quad F = \begin{bmatrix} \rho_1 & \rho_2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad Q = \text{Var}(w_t) = \begin{bmatrix} \sigma_u^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma_\varepsilon^2 \end{bmatrix}. \quad (14)$$

Stacking the firm-count observation with the inflation equation net of its known part,

$$y_t^\pi = y_t - \alpha a_t - \kappa_0 x_t - \lambda_{e\zeta} \zeta_t \quad (15)$$

gives

$$\underbrace{\begin{bmatrix} N_t^{\text{obs}} \\ y_t^\pi \end{bmatrix}}_{z_t} = \underbrace{\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & \delta x_t \end{bmatrix}}_{H_t} s_t + v_t, \quad \text{Var}(v_t) = \begin{bmatrix} \sigma_N^2 & 0 \\ 0 & \sigma_\eta^2 \end{bmatrix}. \quad (16)$$

Conditional on the coefficients and variances this is a linear-Gaussian state-space model, so $s_{0:T}$ can be drawn jointly by an exact Kalman filter / forward-filtering backward-sampling (FFBS).

The initial state starts from the fixed prior

$$s_0 = (\hat{N}_0, \hat{N}_{-1}, \bar{N}_0)' \sim N(m_0, P_0), \quad m_0 = (0, 0, 0)', \quad P_0 = 10I_3 \quad (17)$$

We do not substitute N_0^{obs} into m_0 ; the period-0 firm count and inflation enter the ordinary Kalman observation update. The first AR(2) transition also actually uses the prior-updated $\hat{N}_{-1}$. This avoids double-using the initial observation or dropping the first AR(2) likelihood.

3.4 Conditional posteriors of the coefficients and variances

Take the main model as an example, with $\beta = (\alpha, \kappa_0, \delta)'$, $X_t = (a_t, x_t, x_t \bar{N}_t)$, and $y_t^* = y_t - \lambda_{e\zeta} \zeta_t$. Under the independent normal prior $\beta \sim N(b_0, V_0)$, we update jointly from

$$V_1 = \left(V_0^{-1} + \frac{X'X}{\sigma_\eta^2} \right)^{-1}, \quad (18)$$

$$b_1 = V_1 \left(V_0^{-1} b_0 + \frac{X'y^*}{\sigma_\eta^2} \right), \quad (19)$$

$$\beta \mid \text{rest} \sim N(b_1, V_1) \quad (20)$$

The implementation uses the 100-rescaled X and coefficients noted above, but the formula is identical. The baseline estimation imposes no sign restriction, so both δ and the end-of-sample κ_t can be negative.

We update $\lambda_{e\zeta}$ from the normal conditional posterior of the regression of e_t on ζ_t . We update ϕ_1 using not only (7) but also the information in $e_t = \lambda_{e\zeta}(x_t - \phi_1 x_{t-1}) + \eta_t$. We draw ρ_1, ρ_2 from a normal conditional posterior truncated to the AR(2) stationary region,

$$|\rho_2| < 1, \quad \rho_1 + \rho_2 < 1, \quad \rho_2 - \rho_1 < 1 \quad (21)$$

The fallback count and rejection rate when no proposal is accepted within 2,000 draws are saved to each run's metadata.

The variance priors are $\sigma^2 \sim \text{IG}(a, b)$ (density kernel $(\sigma^2)^{-(a+1)} \exp(-b/\sigma^2)$), updated from the inverse-gamma conditional posterior that adds the residual sum of squares. The concrete baseline priors are in Table 2.

Table 2: Baseline priors for the main HSA steady model. Normal distributions show mean and standard deviation; inverse-gamma distributions show (a, b) . The coefficient priors are common to the other models.

Block	Parameter	Prior	Interpretation
Inflation eq.	α	$N(0.5, 0.2^2)$	Backward inertia
	κ, κ_0	$N(0.1, 0.2^2)$	Slope level
	δ	$N(0, 0.02^2)$	Slope dependence on firm count
	θ, θ_0	$N(0.1, 0.2^2)$	Cyclical entry effect
	γ	$N(0, 0.02^2)$	Time-varying coefficient of entry
Activity eq.	ϕ_1	$N(0.7, 0.2^2)$	AR(1) of the activity variable
	$\lambda_{e\zeta}$	$N(0, 0.5^2)$	Simultaneous-shock correction
Firm-count state	(ρ_1, ρ_2)	$N(0.5, 0.2^2) \times N(-0.5, 0.2^2)$	Truncated to stationary region
	n	$N(0, 0.1^2)$	Trend drift
Variances	$\sigma_\eta^2, \sigma_\zeta^2$	$IG(2, 2), IG(.001, .001)$	Inflation / activity shock
	$\sigma_u^2, \sigma_\varepsilon^2, \sigma_N^2$	$IG(2, .02), IG(2, .01), IG(2, .01)$	State / measurement error

For HSA dynamic the shock covariance takes $\Sigma \sim IW(8, S_0)$ with $S_0 = \text{diag}(3, 3, 0.06, 0.03)$. Under the default (e, ζ) -only, the corresponding marginals of this prior apply to the (e, ζ) block and the two state variances.

3.5 What the Kalman filter and FFBS do

Write the predicted mean and variance at time t as $a_t^s, P_{t|t-1}$ and the filtered mean and variance as m_t, P_t . The forward update for (14)–(16) is

$$a_t^s = c + Fm_{t-1}, \quad P_{t|t-1} = FP_{t-1}F' + Q, \quad (22)$$

$$q_t = z_t - H_t a_t^s, \quad S_t = H_t P_{t|t-1} H_t' + R, \quad (23)$$

$$K_t = P_{t|t-1} H_t' S_t^{-1}, \quad m_t = a_t^s + K_t q_t \quad (24)$$

The covariance update uses the numerically stable Joseph form. After drawing the endpoint $s_T \sim N(m_T, P_T)$, we draw the smoothed state path jointly with

$$J_t = P_t F' P_{t+1|t}^{-1}, \quad (25)$$

$$s_t \mid s_{t+1}, z_{1:T} \sim N(m_t + J_t(s_{t+1} - c - Fm_t), P_t - J_t P_{t+1|t} J_t') \quad (26)$$

for $t = T - 1, \dots, 0$. Thus the plotted $\bar{N}_t, \hat{N}_t, \kappa_t$ carry the posterior uncertainty not only of the coefficients but also of the state decomposition.

3.6 One Gibbs iteration and the differences across models

One iteration of HSA steady is, in order:

1. update $(\alpha, \kappa_0, \delta)$ from the current $\bar{N}_t$ via (20);
2. update $\lambda_{e\zeta}$ and ϕ_1 , and recompute ζ_t, η_t ;
3. update $\sigma_\zeta^2, \sigma_\eta^2$;
4. update (ρ_1, ρ_2) and σ_u^2 ;
5. update n and σ_ε^2 , then σ_N^2 ;
6. update the entire $s_{0:T}$ by FFBS using the updated coefficients and variances.

CES is a conjugate regression benchmark with no latent firm-count state.

HSA dynamic handles the covariance of $r_t = (e_t, \zeta_t, u_t, \varepsilon_t)'$; under the default (e, ζ) -only it updates the 2×2 block of (e_t, ζ_t) and $\sigma_u^2, \sigma_\varepsilon^2$ directly on the constrained parameter space, rather than zeroing the off-diagonal of an unconstrained inverse-Wishart draw after the fact.

HSA full contains the state-by-state product $-\gamma \bar{N}_t \hat{N}_t$, so the joint firm-count state $(\bar{N}_t, \hat{N}_t)$ is not linear-Gaussian at once. In this edition HSA full is therefore updated by the joint *Particle Gibbs* step detailed in Section 3.7—a conditional sequential-Monte-Carlo update of the whole state path (512 particles)—which replaces the two alternating exact-FFBS blocks used previously. HSA const-theta ($\gamma = 0$) still uses those two alternating conditional FFBS blocks: $\hat{N}_{0:T} \mid \bar{N}_{0:T}$, rest by AR(2) FFBS and $\bar{N}_{0:T} \mid \hat{N}_{0:T}$, rest by random-walk FFBS in turn. In both models the weak signal of the cyclical state $\hat{N}_t$, which must be identified as a separate coefficient, keeps convergence weaker than for the other models; the joint Particle Gibbs update improves the mixing of the HSA full coefficients but does not remove this weak identification.

3.7 The Particle Gibbs state update (HSA full)

For HSA full the bilinear term $-\gamma \bar{N}_t \hat{N}_t$ in (6) makes the joint state $s_t = (\hat{N}_t, \hat{N}_{t-1}, \bar{N}_t)'$ non-linear-Gaussian, so the exact FFBS blocks used above do not apply jointly. We instead draw the whole path $s_{0:T}$ with one *Particle Gibbs* sweep—a conditional sequential-Monte-Carlo step that leaves the exact conditional $p(s_{0:T} \mid \theta, y, N^{\text{obs}})$ invariant for any number of particles P (Andrieu, Doucet and Holenstein, 2010).

Proposal and weights. Particles are propagated from the state transition (10)–(11) (a bootstrap proposal): given the ancestor $s_{t-1}^{(i)}$,

$$\hat{N}_t^{(i)} = \rho_1 \hat{N}_{t-1}^{(i)} + \rho_2 \hat{N}_{t-2}^{(i)} + u_t^{(i)}, \quad \bar{N}_t^{(i)} = n + \bar{N}_{t-1}^{(i)} + \varepsilon_t^{(i)}, \quad u_t^{(i)} \sim N(0, \sigma_u^2), \quad \varepsilon_t^{(i)} \sim N(0, \sigma_\varepsilon^2). \quad (27)$$

Each particle is weighted by *both* observation equations—the inflation equation (6) and the firm-count measurement (9):

$$\log w_t^{(i)} = -\frac{(y_t - \mu_t^{(i)})^2}{2\sigma_\eta^2} - \frac{(N_t^{\text{obs}} - \bar{N}_t^{(i)} - \hat{N}_t^{(i)})^2}{2\sigma_N^2}, \quad (28)$$

$$\mu_t^{(i)} = \alpha a_t + (\kappa_0 + \delta \bar{N}_t^{(i)})x_t - (\theta_0 + \gamma \bar{N}_t^{(i)})\hat{N}_t^{(i)} + \lambda_{e\zeta}\zeta_t, \quad (29)$$

with $y_t = \pi_t - \mathbb{E}_t \pi_{t+1}$, $a_t = \pi_{t-1} - \mathbb{E}_t \pi_{t+1}$ and $\zeta_t = x_t - \phi_1 x_{t-1}$ (constants common to all particles at t cancel under normalisation). Normalised weights $W_t^{(i)} = w_t^{(i)} / \sum_j w_t^{(j)}$ are formed with a log-sum-exp for stability, and $\text{ESS}_t = 1 / \sum_i (W_t^{(i)})^2$.

Conditional SMC. Let $s_{0:T}^{\text{ref}}$ be the state path from the previous MCMC iteration, pinned to particle slot 1.

1. $t = 0$: set $s_0^{(1)} = s_0^{\text{ref}}$ and draw $s_0^{(i)} \sim N(m_0, P_0)$ for $i \geq 2$ from the initial prior (17); weight by (28).
2. $t = 1, \dots, T$: resample ancestors $a_t^{(i)} \sim \text{Categorical}(W_{t-1})$ for $i \geq 2$ and set $a_t^{(1)} = 1$, so the reference keeps its lineage; propagate $s_t^{(i)}$ from the transition above given $s_{t-1}^{(a_t^{(i)})}$, with $s_t^{(1)} = s_t^{\text{ref}}$; weight by (28). The pair $(\hat{N}_t, \hat{N}_{t-1})$ in s_t preserves the AR(2) genealogy.
3. Draw a terminal index $J \sim \text{Categorical}(W_T)$ and trace its ancestors backward to return one joint path $s_{0:T}$, which is the new state draw and the reference for the next iteration.

(section 9).

3.8 Iterations, saved draws, and convergence criteria

Each cell uses two independent chains, with a *total* of 12,000 iterations per chain. We discard the first 4,000 as burn-in and save every 5th of the remaining 8,000, giving 1,600 draws per chain and 3,200 draws per cell for diagnostics and summaries. The main analysis is 75 cells: the baseline 45 (3 price indices $\times$ 3 activity measures $\times$ 5 models) plus the 30 weak/tight-prior cells. Separately, we estimate 2 inverse-markup CES/HSA-dynamic cells with the same settings.

The acceptance criteria are maximum $\hat{R} \leq 1.01$ and minimum bulk ESS ≥ 400 . HSA steady and CES pass for all reported cells. Many const-theta/full cells and some dynamic cells fail; they are marked $\dagger$ in the coefficient tables and dropped from the main inference. The 95% credible intervals are the 2.5% and 97.5% quantiles of the saved draws; the plotted points are posterior means.

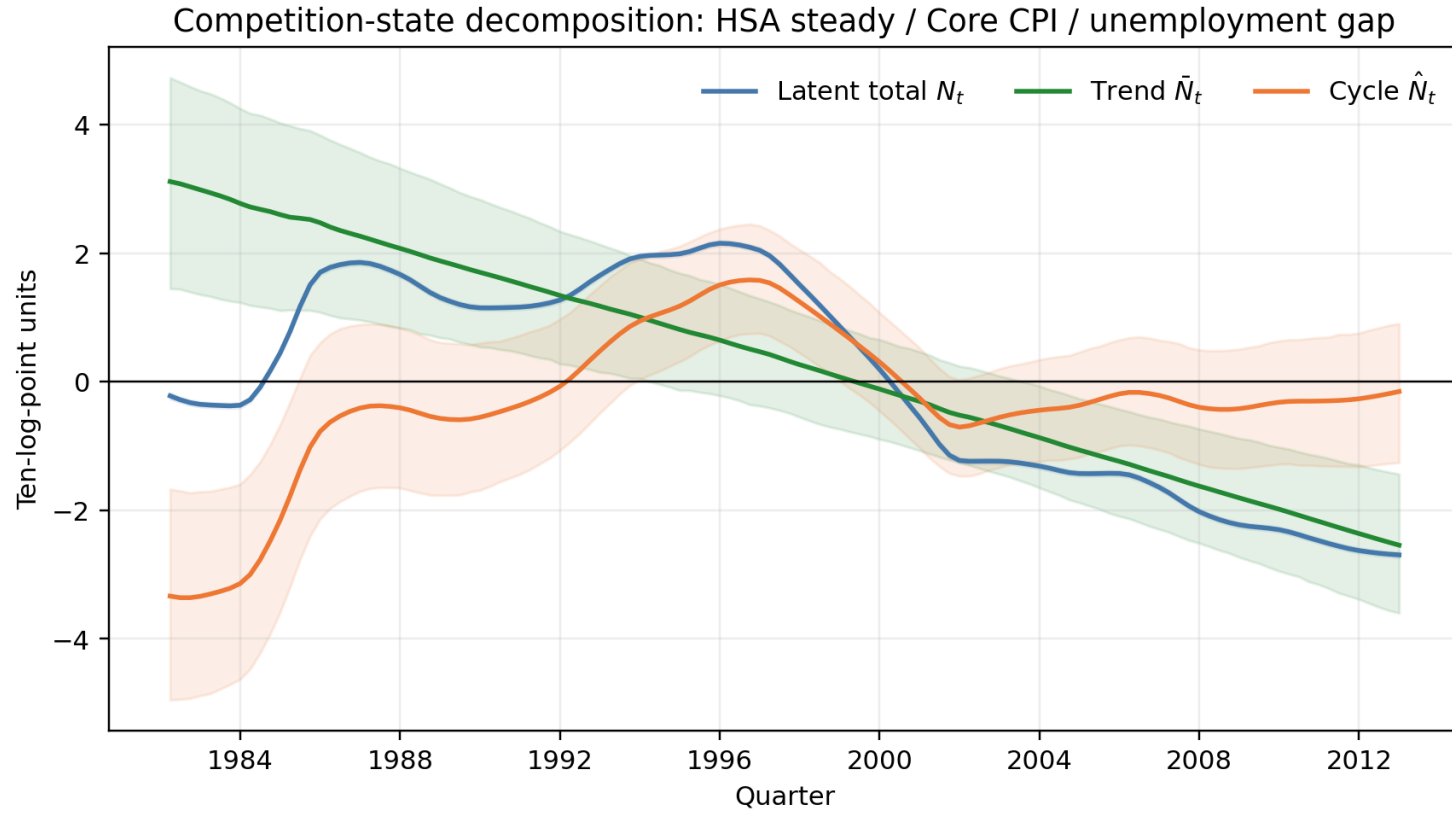


Figure 2: Posterior decomposition of the firm count $N_t = \bar{N}_t + \hat{N}_t$ under the main HSA steady (negative unemployment gap, core CPI). Lines are posterior means, bands are 95% credible intervals. That the state paths are smoothly estimated does not by itself imply identification of κ_0 or δ .

4 Main results

Table 3 gives the main-model (HSA steady) estimates for every price index and activity measure, so each result can be read against the specification it comes from. The main specification—**core CPI, negative unemployment gap**—is in bold; a positive δ means the slope flattens as the firm count falls.

Table 3: Headline results (main model: HSA steady, baseline priors, PCHIP, $T = 124$), by price index and activity measure. δ is the competition dependence of the slope (a positive δ means the slope flattens as the firm count falls); BF_{10} is the Savage–Dickey Bayes factor against $\delta = 0$; κ_0 is the slope at average competition; the last column is the implied κ_t from the start to the end of the sample. The **main specification (core CPI, negative unemployment gap)** is in bold. †: outside the convergence criteria ($\hat{R} \leq 1.01$, bulk ESS ≥ 400).

Activity measure	Price	δ [95% CI]	BF_{10}	κ_0 [95% CI]	κ_t start→end	Conv.
Negative unemployment gap	Core CPI	+0.023 [+0.010 , +0.039]	429.7	+0.051 [+0.016 , +0.090]	+0.120 → -0.005	OK
	Headline CPI	+0.030 [+0.008, +0.051]	22.8	+0.067 [+0.002, +0.136]	+0.163 → -0.010	OK
	PPI	+0.012 [-0.024, +0.049]	1.2	+0.075 [-0.131, +0.279]	+0.113 → +0.044	OK
HP output gap	Core CPI	+0.029 [+0.011, +0.048]	37.3	+0.069 [+0.018, +0.123]	+0.161 → -0.005	OK
	Headline CPI	-0.002 [-0.031, +0.028]	0.8	+0.141 [+0.048, +0.237]	+0.135 → +0.146	OK
	PPI	-0.005 [-0.044, +0.034]	1.0	+0.210 [-0.067, +0.473]	+0.193 → +0.224	OK
BN output gap	Core CPI	+0.011 [-0.008, +0.031]	1.0	+0.031 [-0.020, +0.083]	+0.063 → +0.005	OK
	Headline CPI	-0.013 [-0.042, +0.016]	1.1	+0.064 [-0.024, +0.153]	+0.022 → +0.097	OK
	PPI	-0.007 [-0.044, +0.030]	1.1	+0.190 [-0.054, +0.427]	+0.167 → +0.210	OK

Table 4 then places the main model against the other model variants (CES, HSA dynamic, HSA const-theta, HSA full) for the negative unemployment gap. CES and HSA dynamic hold the slope constant, so they have no δ ; the HSA models with a firm-count-dependent slope estimate δ , and HSA steady is the main model. Full coefficients (θ , γ , and the κ_t path) for every activity measure are in Appendix C.

Table 4: Model comparison for the negative unemployment gap (baseline priors, PCHIP), by price index. Slope is κ for CES / HSA dynamic and κ_0 otherwise; the competition dependence δ (and its BF_{10}) is defined only for the HSA models with a firm-count-dependent slope. The main model, **HSA steady**, is in bold; full coefficients (θ , γ , κ_t path) for every activity measure are in Appendix C. †: outside the convergence criteria.

Model	Price	slope [95% CI]	δ [95% CI]	$\text{BF}_{10}(\delta)$	Conv.
CES	Core CPI	+0.040 [+0.007, +0.071]	–	–	OK
	Headline CPI	+0.057 [-0.006, +0.120]	–	–	OK
	PPI	+0.073 [-0.145, +0.284]	–	–	OK
HSA steady	Core CPI	+0.051 [+0.016, +0.090]	+0.023 [+0.010, +0.039]	429.7	OK
	Headline CPI	+0.067 [+0.002, +0.136]	+0.030 [+0.008, +0.051]	22.8	OK
	PPI	+0.075 [-0.131, +0.279]	+0.012 [-0.024, +0.049]	1.2	OK
HSA dynamic	Core CPI	+0.032 [-0.004, +0.066]	–	–	OK
	Headline CPI	+0.052 [-0.026, +0.143]	–	–	watch
	PPI	+0.057 [-0.153, +0.276]	–	–	OK
HSA const-theta	Core CPI	+0.063 [+0.024, +0.101]	+0.031 [+0.012, +0.050] [†]	49.8	watch
	Headline CPI	+0.089 [+0.017, +0.163]	+0.030 [-0.008, +0.062] [†]	4.1	watch
	PPI	+0.082 [-0.142, +0.304]	+0.011 [-0.026, +0.048] [†]	1.1	watch
HSA full	Core CPI	+0.061 [+0.023, +0.097]	+0.028 [+0.008, +0.053] [†]	20.3	watch
	Headline CPI	+0.082 [+0.004, +0.156]	+0.032 [+0.004, +0.060]	7.8	OK
	PPI	+0.087 [-0.130, +0.307]	+0.011 [-0.027, +0.049] [†]	1.2	watch

4.1 The negative-unemployment-gap slope flattens as the firm count falls

For the baseline HSA steady on the main sample, the core-CPI negative-unemployment-gap specification gives

$$\delta = +0.023[+0.010, +0.039],$$

and the Savage–Dickey Bayes factor against $\delta = 0$ is $\text{BF}_{10} = 429.7$. Because $\bar{N}_t$ falls over the sample, the implied slope κ_t declines monotonically from +0.120 to –0.005: as the firm count falls (concentration rises), the Phillips curve flattens, the sign the HSA mechanism predicts. Headline CPI also gives $\delta = +0.030[+0.008, +0.051]$ ($\text{BF}_{10} = 22.8$), so the CPI result for the negative unemployment gap is robust to the inflation index. Figure 3 plots the implied κ_t path; its end-of-sample band includes zero, so the accurate statement is a flattening to near zero rather than a strictly positive residual slope.

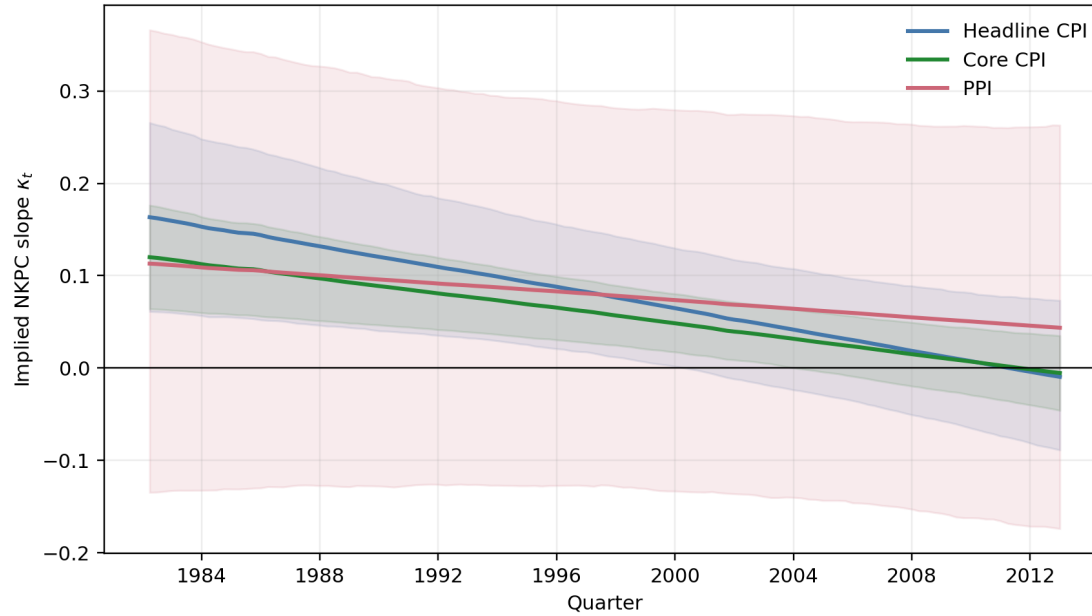


Figure 3: Time-varying slope κ_t for the negative unemployment gap (posterior mean and 95% band). Under CPI it declines to near zero by the end of the sample, whereas the PPI band is wide.

The estimated Phillips-curve slope in the core-CPI specification declines from about 0.12 at the beginning of the sample to approximately zero by the end of the sample. This change is economically meaningful. For example, a four-point deterioration in the negative unemployment gap ($x_t = -4$) implies an inflation response of about -0.48 percentage point at the 1982 slope, but almost no response at the 2012 slope.

5 The CES slope bias and the constant-slope benchmark

5.1 Theoretical prediction and its empirical counterpart

Fujiwara–Matsuyama (2026) show that because endogenous entry generates a cost-push term through strategic complementarity, a CES-type equation that regresses inflation on real marginal cost alone under-estimates the NKPC slope. In our HSA-dynamic representation, moving expected inflation to the left-hand side gives

$$y_t = \alpha a_t + \kappa_{\text{HSA}} x_t + q_t + e_t, \quad q_t = -\theta \hat{N}_t, \quad (30)$$

where $y_t = \pi_t - \mathbb{E}_t \pi_{t+1}$ and $a_t = \pi_{t-1} - \mathbb{E}_t \pi_{t+1}$. When the CES equation omits q_t ,

$$\text{Bias}(\hat{\kappa}_{\text{CES}}) = \text{plim}(\hat{\kappa}_{\text{CES}} - \hat{\kappa}_{\text{HSA}}) = \frac{\tilde{x}' \tilde{q}}{\tilde{x}' \tilde{x}}, \quad (31)$$

where tildes denote residuals after removing a_t and the activity shock $\zeta_t = x_t - \phi_1 x_{t-1}$. The theory predicts (31) is negative, i.e. $\kappa_{\text{CES}} - \kappa_{\text{HSA}} < 0$.

To keep this correspondence sharp, separately from the slack measures of the main results we use the BN-decomposed inverse markup as a proxy for real marginal cost and re-estimate CES and HSA dynamic with the same sample, priors, and MCMC settings. For each HSA posterior draw we form $q_t^{(s)} = -\theta^{(s)} \hat{N}_t^{(s)}$ and directly compute the bias in (31). Because the CES–HSA difference is a comparison across separately estimated posteriors, we build a Monte Carlo distribution of the difference by randomly permuting the HSA draws with a fixed seed.

5.2 Estimation results

For the inverse-markup specification that corresponds directly to the theory,

$$\begin{aligned} \kappa_{\text{CES}} &= +0.007[-0.217, +0.228], & \kappa_{\text{HSA}} &= -0.011[-0.226, +0.215], \\ \kappa_{\text{CES}} - \kappa_{\text{HSA}} &= +0.018[-0.291, +0.323], & \Pr(\kappa_{\text{CES}} - \kappa_{\text{HSA}} < 0) &= 0.456, \\ B_{\text{HSA}} &= +0.022[-0.024, +0.060], & \Pr(B_{\text{HSA}} < 0) &= 0.072 \end{aligned}$$

The CES–HSA difference does not show a negative bias. The omitted-variable bias implied by HSA itself also has a point estimate of the sign opposite to the theory (positive), with a credible interval that includes zero.

Table 5: CES–HSA-dynamic slope comparison and the omitted-variable bias implied by HSA. Each interval is a 95% credible interval. † marks that at least one of CES or HSA dynamic is outside the convergence criteria.

specification	CES kappa	HSA kappa	CES-HSA	Pr(CES-HSA _i 0)	HSA-implied OVB	Pr(OVB _i 0)	diagnostics
Inverse markup (theory)	+0.007 [-0.217, +0.228]	-0.011 [-0.226, +0.215]	+0.018 [-0.291, +0.323]	0.456	+0.022 [-0.024, +0.060]	0.072	OK
Headline CPI / negative unemployment gap	+0.057 [-0.006, +0.120] [†]	+0.052 [-0.026, +0.143] [†]	+0.005 [-0.101, +0.105] [†]	0.452	+0.004 [-0.064, +0.052] [†]	0.371	watch [†]
Core CPI / negative unemployment gap	+0.040 [+0.007, +0.071]	+0.032 [-0.004, +0.066]	+0.007 [-0.040, +0.054]	0.388	+0.007 [-0.010, +0.027]	0.207	OK
PPI / negative unemployment gap	+0.073 [-0.145, +0.284]	+0.057 [-0.153, +0.276]	+0.016 [-0.288, +0.317]	0.468	-0.000 [-0.111, +0.104]	0.507	OK

Verdict. The inverse-markup row passes the convergence criteria for both models, but the slope intervals are wide and neither the CES–HSA difference nor the within-model OVB supports the negative sign the theory predicts. The three auxiliary rows using the negative unemployment gap give the same conclusion. This does not amount to a refutation of the theory: the inverse markup is an aggregate series rather than firm-level real marginal cost, and HSA dynamic is not a joint estimate of the full theoretical model. What we can say from this paper is only that *in the current data and semi-structural model, we do not obtain an estimate corresponding to the theory paper’s prediction of a negative bias in the CES slope.*

As a benchmark for the time-varying result of Section 4, we also estimate the constant-slope models. The full model-by-model coefficient tables for the negative unemployment gap, and the corresponding prior-sensitivity table, are collected in Appendix C.

CES vs HSA (constant κ) CES is the constant-slope benchmark. Among the converged cells, the core-CPI κ interval sits on the positive side, while headline CPI and PPI include zero. CES measures the existence of the average curve but does not explain the flattening. The prior-vs-posterior of the slope (Figure 4) shows CES’s constant κ and HSA steady’s slope at average competition κ_0 , both sharply updated by the data and sitting just above zero.

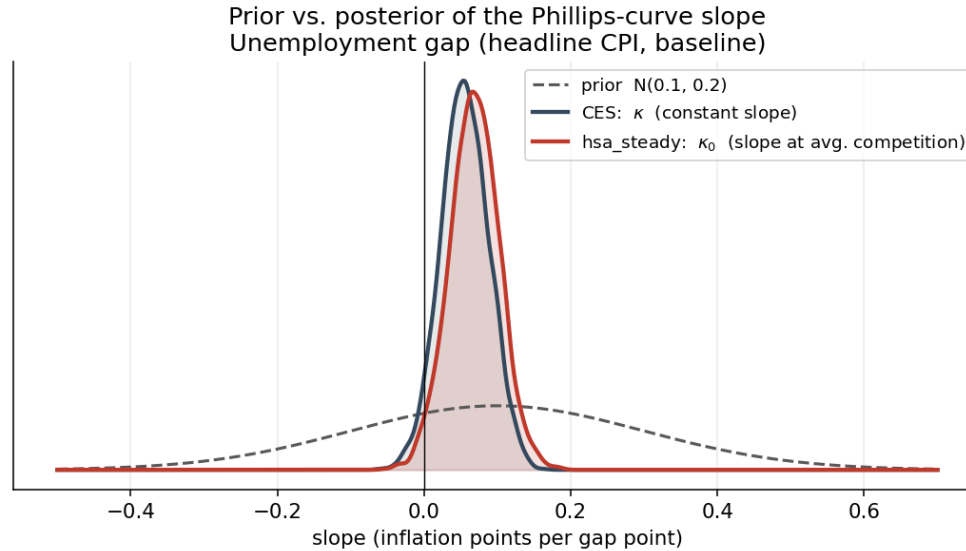


Figure 4: Prior vs. posterior of the Phillips-curve slope for the negative unemployment gap: CES’s constant slope κ and HSA steady’s slope at average competition κ_0 , against their common prior $\mathcal{N}(0.1, 0.2)$. Both are sharply updated by the data and sit just above zero; the two models agree on the slope level, so what distinguishes them is whether that slope varies with the firm count (Section 6).

6 Does the time-varying slope improve fit over the constant slope?

We compare CES (constant κ) against HSA steady (slope varying with the firm count) by two criteria: an in-sample fit score and the Savage–Dickey Bayes factor for $\delta \neq 0$ (Table 6).

In-sample plug-in fit score. We evaluate the inflation equation at posterior-mean parameters and states (bars denote posterior means),

$$\hat{\pi}_t = \bar{\alpha} \pi_{t-1} + (1 - \bar{\alpha}) \mathbb{E}_t \pi_{t+1} + \bar{\kappa}_t x_t, \quad \bar{\kappa}_t = \bar{\kappa}_0 + \bar{\delta} \bar{N}_t \quad (\text{CES: } \bar{\kappa}_t = \bar{\kappa}), \quad (32)$$

and score the fit by the Gaussian log-likelihood of the residuals $r_t = \pi_t - \hat{\pi}_t$ with $\bar{\sigma}_e^2$ the posterior-mean shock variance:

$$S = -\frac{T}{2} \log(2\pi\bar{\sigma}_e^2) - \frac{1}{2\bar{\sigma}_e^2} \sum_{t=1}^T r_t^2. \quad (33)$$

This is an *in-sample plug-in* score: it reuses the estimation sample and the posterior means, and is neither an out-of-sample forecast nor a posterior predictive density. It is used only as a rough comparative fit diagnostic; a proper predictive comparison (LOO, WAIC, rolling one-step forecasts, or an integrated log predictive density) is left for future work. In the negative-unemployment-gap specification the time-varying slope improves this score over CES.

Bayes factor (Savage–Dickey), stated precisely. We test $\delta = 0$ *within the HSA state model*: M_1 is HSA steady with δ free and M_0 is the same model with $\delta = 0$. The Savage–Dickey identity gives

$$\text{BF}_{10} = \frac{p(\delta = 0)}{\pi(\delta = 0 | y)}, \quad p(\delta = 0) = \frac{1}{\sigma_\delta \sqrt{2\pi}} = \frac{1}{0.02\sqrt{2\pi}} \approx 19.95 \quad (\delta \sim \mathcal{N}(0, \sigma_\delta^2)), \quad (34)$$

with the posterior ordinate estimated by a Gaussian kernel density on the draws $\{\delta^{(s)}\}$, $\hat{\pi}(\delta = 0 | y) = \frac{1}{Sh} \sum_s \phi((0 - \delta^{(s)})/h)$. Three cautions. (i) This is *not* a CES-vs-HSA marginal-likelihood ratio: CES and HSA also differ in the latent- N block, which the SDDR holds fixed, so we keep the comparison inside the HSA state model. (ii) The theory is one-sided ($\delta > 0$); the two-sided BF_{10} is conservative. (iii) The zero-point posterior density is a KDE estimate; the ordering across specifications is insensitive to the bandwidth, but the exact magnitudes are not, so we report them to a few significant figures.

Table 6: Constant vs. time-varying slope: is $\delta \neq 0$? Posterior mean of δ [95% CI] with the Savage–Dickey Bayes factor BF_{10} for $\delta \neq 0$ within the HSA state model in parentheses (HSA steady, baseline priors, quarterly, unrestricted). Values are auto-generated from the current posteriors; $\text{BF}_{10} > 1$ favors a concentration-dependent slope.

Activity measure	Headline CPI	Core CPI	PPI
Negative unemployment gap	+0.030 [+0.008, +0.051] (22.8)	+0.023 [+0.010, +0.039] (429.7)	+0.012 [-0.024, +0.049] (1.2)
Output gap (HP)	-0.002 [-0.031, +0.028] (0.8)	+0.029 [+0.011, +0.048] (37.3)	-0.005 [-0.044, +0.034] (1.0)
Output gap (BN)	-0.013 [-0.042, +0.016] (1.1)	+0.011 [-0.008, +0.031] (1.0)	-0.007 [-0.044, +0.030] (1.1)

Verdict. For the negative unemployment gap, allowing the slope to vary with the firm count improves the in-sample fit score and the Bayes factor favors $\delta \neq 0$ strongly under both CPI indices. For the headline output gaps the gain is marginal, consistent with the appendix tables: a positive competition dependence appears only in the core-CPI HP specification, not in BN. For the labor-share gap and inverse markup there is no inflation signal, so constant and time-varying slopes fit equally well. We stress that this is an in-sample fit comparison, not evidence of out-of-sample predictive gains.

Limits of the model comparison. We read the Savage–Dickey ratios only as the point null $\delta = 0$ *within the HSA state model*; they are not a CES-vs-HSA marginal-likelihood ratio. The in-sample LPPD and RMSE conditioned on the latent states are likewise not out-of-sample predictions, so they are not used as grounds to select a more complex model. The automatic Chib marginal likelihood is contaminated for annual-Q4 blocks by a missing-N bug (Section 11) and is excluded from this table.

7 Estimating quarterly firm counts from Q4-only observations

Against the results so far, which use the PCHIP-interpolated value as data every quarter, this section applies the mixed-frequency observation scheme of (12) and reports the results in the same order. The 3 price indices, 3 activity measures, 5 models, 3 priors, and the inverse-markup slope-bias diagnostic are unchanged. We newly estimate the 61 HSA cells with 12,000 iterations and 2 chains. Because CES contains neither N_t nor a latent firm-count state, its likelihood and posterior do not change with the observation frequency, so we reuse it as a shared reference across the 16 CES cells. In the CES–HSA comparisons below, the CES column is therefore the same benchmark and only the HSA column is annual-Q4 estimated.

7.1 What the annual-Q4 estimation does

The annual-Q4 variant is a *mixed-frequency state-space estimation*, not an interpolate-then-regress procedure. The distinction is the whole point: the PCHIP version inserts a smooth deterministic curve and treats it as if the firm count were observed every quarter, which injects manufactured within-year variation and understates uncertainty; the annual-Q4 version instead treats the sub-annual firm count as *latent* and lets the model infer it.

Concretely, the firm count enters the observation equation only in the 31 fourth quarters, and Q1–Q3 are coded as missing ((12)). The Kalman filter handles this exactly by varying the observation block by quarter. In a Q4 the update uses both rows of (16)—the firm-count row and the inflation row—so the annual anchor pins the level of $\bar{N}_t$. In Q1–Q3 the firm-count row is dropped and only the inflation row remains, i.e. the measurement update for that quarter carries just the information that the slope δx_t places on $\bar{N}_t$; the state prediction and transition (14) nonetheless run at every quarter, propagating the latent $(\hat{N}_t, \hat{N}_{t-1}, \bar{N}_t)$ forward between the annual observations. Each Gibbs iteration then draws the *entire* quarterly path $s_{0:T}$ by FFBS, conditioned on the 31 annual anchors, the quarterly inflation and activity data, the state dynamics, and the current coefficients and variances. The Q1–Q3 firm counts are therefore posterior draws, not deterministic interpolations, and their uncertainty flows through to the posterior of δ and the κ_t path.

The mixed-frequency Kalman step, in equations. Yes, the state block is an exact Kalman filter—the same one as in Section 3; annual-Q4 changes only the observation row that is available each quarter. Let $\mathcal{T}_N$ be the 31 Q4 quarters and, as before, write the inflation observation net of its known part as $y_t^\pi = y_t - \alpha a_t - \kappa_0 x_t - \lambda_{e\zeta} \zeta_t$. The observation triple (z_t, H_t, R_t) switches with the calendar quarter:

$$(z_t, H_t, R_t) = \begin{cases} \left(\begin{bmatrix} N_t^{\text{obs}} \\ y_t^\pi \end{bmatrix}, \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & \delta x_t \end{bmatrix}, \begin{bmatrix} \sigma_N^2 & 0 \\ 0 & \sigma_\eta^2 \end{bmatrix} \right), & t \in \mathcal{T}_N \text{ (Q4)}, \\ \left(y_t^\pi, \begin{bmatrix} 0 & 0 & \delta x_t \end{bmatrix}, \sigma_\eta^2 \right), & t \notin \mathcal{T}_N \text{ (Q1–Q3)}. \end{cases} \quad (35)$$

The same linear-Gaussian recursion then runs at *every* quarter, with only the row dimension of the update changing (two rows in Q4, one in Q1–Q3):

$$\begin{aligned}
m_{t|t-1} &= c + Fm_{t-1}, & P_{t|t-1} &= FP_{t-1}F' + Q, \\
v_t &= z_t - H_t m_{t|t-1}, & S_t &= H_t P_{t|t-1} H_t' + R_t, \\
K_t &= P_{t|t-1} H_t' S_t^{-1}, & m_t &= m_{t|t-1} + K_t v_t, \\
P_t &= (I - K_t H_t) P_{t|t-1} \quad (\text{Joseph form}).
\end{aligned}$$

In Q1–Q3 the prediction still advances $\bar{N}_t, \hat{N}_t$ through (14), but the firm-count row is simply absent and the only measurement information is the scalar y_t^π loading on $\bar{N}_t$ through δx_t . The smoothed path $s_{0:T}$ is drawn by the same forward-filtering backward-sampling pass as the PCHIP version. Finally, the measurement-variance update uses only the finite Q4 residuals,

$$r_t^N = N_t^{\text{obs}} - \hat{N}_t - \bar{N}_t \quad (t \in \mathcal{T}_N), \quad \sigma_N^2 \mid \cdot \sim \text{IG}\left(a_N + \frac{|\mathcal{T}_N|}{2}, b_N + \frac{1}{2} \sum_{t \in \mathcal{T}_N} (r_t^N)^2\right), \quad |\mathcal{T}_N| = 31. \quad (36)$$

Setting $\mathcal{T}_N = \{1, \dots, T\}$ (every quarter observed) recovers the PCHIP filter exactly; annual-Q4 is that filter with 93 of the 124 firm-count rows removed.

In short, within the same Gibbs sweep the annual-Q4 estimation jointly updates:

- the coefficients $(\alpha, \kappa_0, \delta, \dots)$ from (20), exactly as in the PCHIP version;
- the shock and measurement variances, with σ_N^2 updated from only the 31 finite Q4 residuals;
- the full latent path $\bar{N}_{0:T}, \hat{N}_{0:T}$ (all 124 quarters, including the 93 unobserved ones) by FFBS.

CES carries no latent firm-count state and no N_t term, so its likelihood and posterior are unchanged by the observation frequency; it is reused as the common reference. The practical cost is that not pretending the quarterly count is observed widens the intervals and lowers the effective sample size for the state block—which is itself the measurement uncertainty that the PCHIP version hides rather than a defect of the estimator.

7.2 CPI and PPI: main results of HSA steady

We first line up all price indices and activity measures for the main HSA steady. For the negative unemployment gap, headline CPI gives $\delta = +0.029[+0.005, +0.053]$ ($\text{BF}_{10} = 10.8$), core CPI gives $\delta = +0.022[+0.008, +0.038]$ ($\text{BF}_{10} = 44.5$), and PPI gives $\delta = +0.009[-0.028, +0.047]$ ($\text{BF}_{10} = 1.1$). The corresponding PCHIP values are, in order, $+0.030 [+0.008, +0.051]$, $+0.023 [+0.010, +0.039]$, and $+0.012 [-0.024, +0.049]$, so the change in point estimate and interval from the change in observation frequency can be compared directly.

Table 7: Annual-Q4 version: HSA-steady results by price index and activity measure. δ is posterior mean [95% CI]; κ path is the change in posterior mean from the start to the end of the sample. † marks failure of the whole-run convergence criteria.

inflation	activity	delta	BF10	kappa path
Headline CPI	Unemployment gap	+0.029 [+0.005, +0.053]†	10.8	+0.143 → -0.001†
Headline CPI	HP output gap	-0.006 [-0.039, +0.027]†	1.0	+0.135 → +0.155†
Headline CPI	BN output gap	-0.019 [-0.052, +0.013]†	1.6	+0.037 → +0.103†
Core CPI	Unemployment gap	+0.022 [+0.008, +0.038]†	44.5	+0.110 → -0.002†
Core CPI	HP output gap	+0.028 [+0.009, +0.047]†	19.5	+0.152 → +0.001†
Core CPI	BN output gap	+0.000 [-0.031, +0.025]†	0.8	+0.041 → +0.030†
PPI	Unemployment gap	+0.009 [-0.028, +0.047]†	1.1	+0.093 → +0.055†
PPI	HP output gap	-0.004 [-0.043, +0.035]†	1.0	+0.207 → +0.221†
PPI	BN output gap	-0.008 [-0.046, +0.031]†	1.0	+0.177 → +0.207†

It is important to distinguish “the marginal chains of the coefficients mixed” from “the joint posterior including the latent states converged.” For the three price indices of the negative unemployment gap, the maximum $\hat{R}$ of δ is 1.000 throughout, the minimum bulk ESS exceeds 3,000, and κ_0 has $\hat{R} \leq 1.003$ and ESS above 1,000. By contrast, the $\tilde{N}_t$ and $\hat{N}_t$ paths have time points with $\hat{R} > 1.01$ or ESS < 400, so the 15 HSA-steady cells fail the whole-run criteria. Table 8 shows this difference without hiding it.

Table 8: Annual-Q4 version: coefficient-by-coefficient / state-by-state diagnostics for HSA steady, negative unemployment gap. Rhat is the maximum along the path, ESS the minimum bulk ESS along the path. The acceptance criteria are Rhat ≤ 1.01 and ESS ≥ 400 .

inflation	delta Rhat	delta ESS	kappa0 Rhat	kappa0 ESS	kappa path Rhat	kappa path ESS	Nbar path Rhat	Nbar path ESS	Nhat path Rhat	Nhat path ESS
Headline CPI	1.000	3378	1.000	2694	1.003	269	1.013	68	1.065	39
Core CPI	1.000	3087	1.003	1076	1.006	153	1.015	74	1.036	67
PPI	1.000	3277	1.003	3068	1.003	2892	1.016	82	1.016	80

Thus the positive sign and magnitude of the annual-Q4 CPI δ are a strong *provisional* reproduction of the PCHIP version, but under this paper’s strict acceptance rule we do not call them a confirmed robustness check. The κ_t paths and state decompositions below are diagnostic figures showing the pointwise-posterior computations, not converged main results.

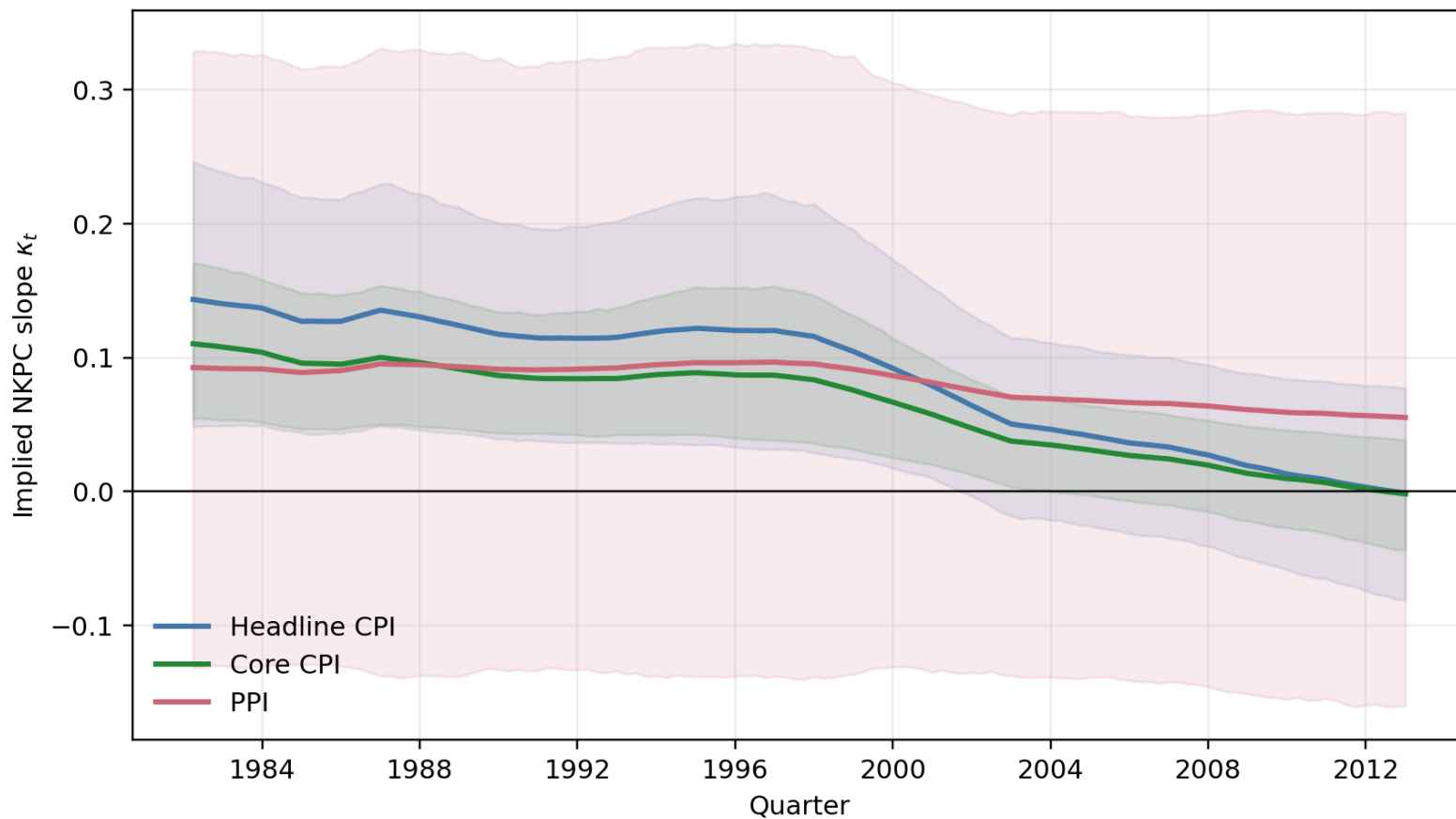


Figure 5: Annual-Q4 version: time-varying slope κ_t for the negative unemployment gap (posterior mean and 95% band). It includes the uncertainty from estimating the Q1–Q3 firm-count states by FFBS, but is provisional because of state-path convergence warnings.

For core CPI and the negative unemployment gap the implied slope moves from $+0.110$ to -0.002 . This path uses a $\bar{N}_t$ extracted jointly from the 31 annual anchors and the quarterly inflation and activity data, not one taken as given from a quarterly PCHIP curve. Figure 6 shows the posterior of the total firm-count state, trend, and cycle, including the unobserved quarters.

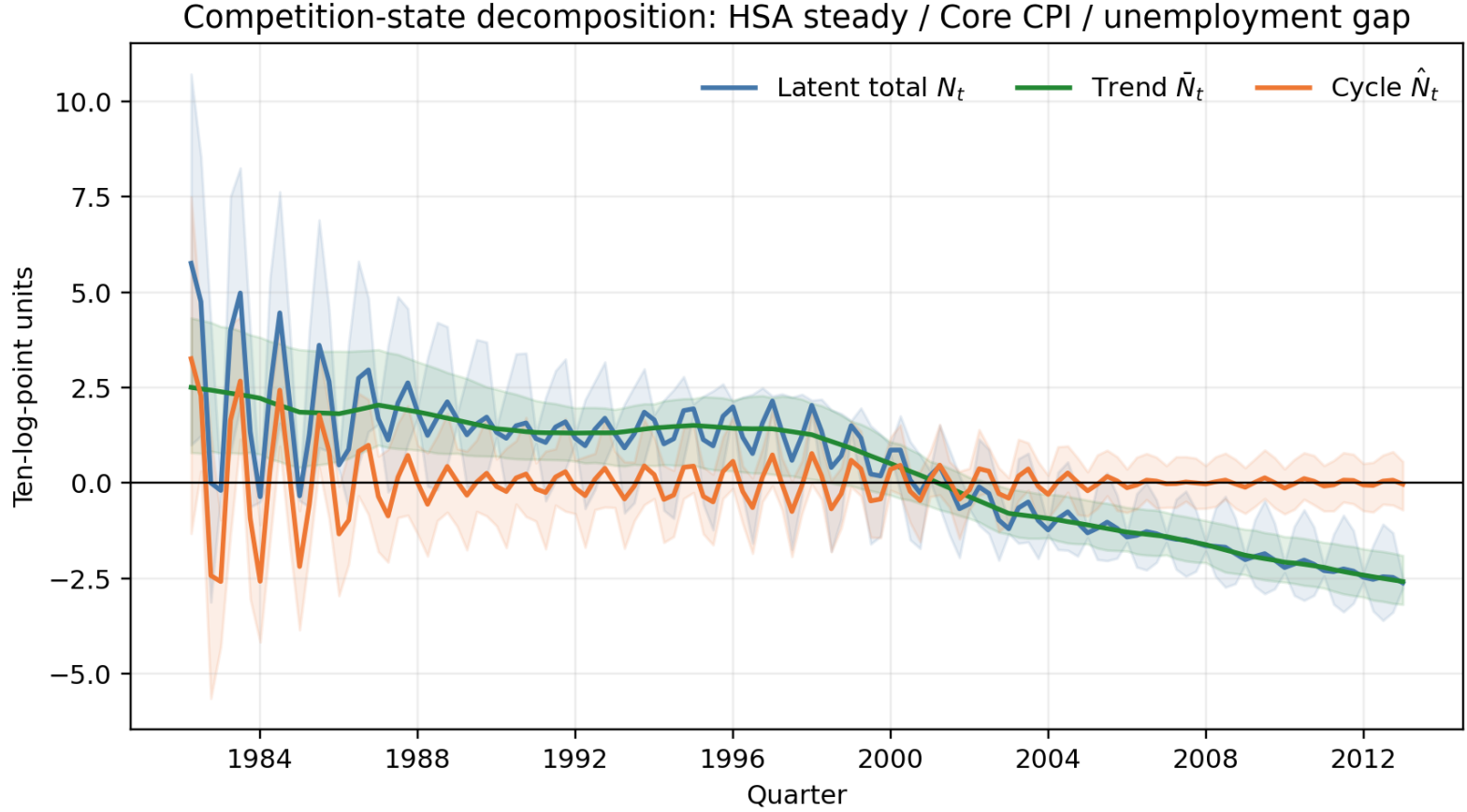


Figure 6: Annual-Q4 version: firm-count state decomposition for core CPI, negative unemployment gap, HSA steady. Lines are posterior means, bands are 95% credible intervals. The quarterly states between the Q4 annual observations are also inferred.

7.3 Comparison of the five models

For the negative unemployment gap we show the same five models as the PCHIP version by price index. Constant-slope CES is the common benchmark; HSA steady is the time-varying slope through $\tilde{N}_t$; HSA dynamic is the entry term in $\hat{N}_t$; const-theta/full combine both channels. The $\dagger$ in the table marks that the annual-Q4 posterior is outside the convergence criteria, and its coefficients are not used for conclusions even when the sign is consistent.

Table 9: Annual-Q4 version: model-by-model results for the negative unemployment gap. CES is a common reference that does not use the firm count.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.057 [-0.006, +0.120]	—	—	—	—	—	OK
CES	Core CPI	+0.040 [+0.007, +0.071]	—	—	—	—	—	OK
CES	PPI	+0.073 [-0.145, +0.284]	—	—	—	—	—	OK
HSA steady	Headline CPI	+0.075 [+0.009, +0.145]	+0.029 [+0.005, +0.053]	10.8	—	—	+0.143 → -0.001	watch [†]
HSA steady	Core CPI	+0.056 [+0.021, +0.092]	+0.022 [+0.008, +0.038]	44.5	—	—	+0.110 → -0.002	watch [†]
HSA steady	PPI	+0.080 [-0.134, +0.296]	+0.009 [-0.028, +0.047]	1.1	—	—	+0.093 → +0.055	watch [†]
HSA dynamic	Headline CPI	+0.055 [-0.011, +0.117]	—	—	+0.037 [-0.211, +0.319]	—	—	watch [†]
HSA dynamic	Core CPI	+0.040 [+0.009, +0.071]	—	—	-0.012 [-0.173, +0.289]	—	—	watch [†]
HSA dynamic	PPI	+0.051 [-0.155, +0.263]	—	—	+0.101 [-0.233, +0.431]	—	—	watch [†]
HSA const-theta	Headline CPI	+0.084 [+0.016, +0.154]	+0.028 [-0.001, +0.055]	4.4	+0.043 [-0.177, +0.317]	—	+0.114 → +0.010	watch [†]
HSA const-theta	Core CPI	+0.058 [+0.024, +0.094]	+0.022 [+0.008, +0.038]	25.6	-0.044 [-0.134, +0.039]	—	+0.108 → -0.001	watch [†]
HSA const-theta	PPI	+0.074 [-0.139, +0.282]	+0.011 [-0.026, +0.048]	1.1	+0.102 [-0.213, +0.404]	—	+0.102 → +0.047	watch [†]
HSA full	Headline CPI	+0.082 [+0.015, +0.151]	+0.028 [+0.002, +0.054]	6.0	+0.029 [-0.166, +0.254]	-0.002 [-0.041, +0.036]	+0.124 → +0.007	watch [†]
HSA full	Core CPI	+0.060 [+0.020, +0.104]	+0.024 [+0.009, +0.041]	21.0	+0.028 [-0.105, +0.170]	-0.009 [-0.041, +0.025]	+0.135 → -0.003	watch [†]
HSA full	PPI	+0.074 [-0.141, +0.290]	+0.009 [-0.028, +0.047]	1.1	+0.109 [-0.218, +0.446]	-0.000 [-0.038, +0.038]	+0.097 → +0.049	watch [†]

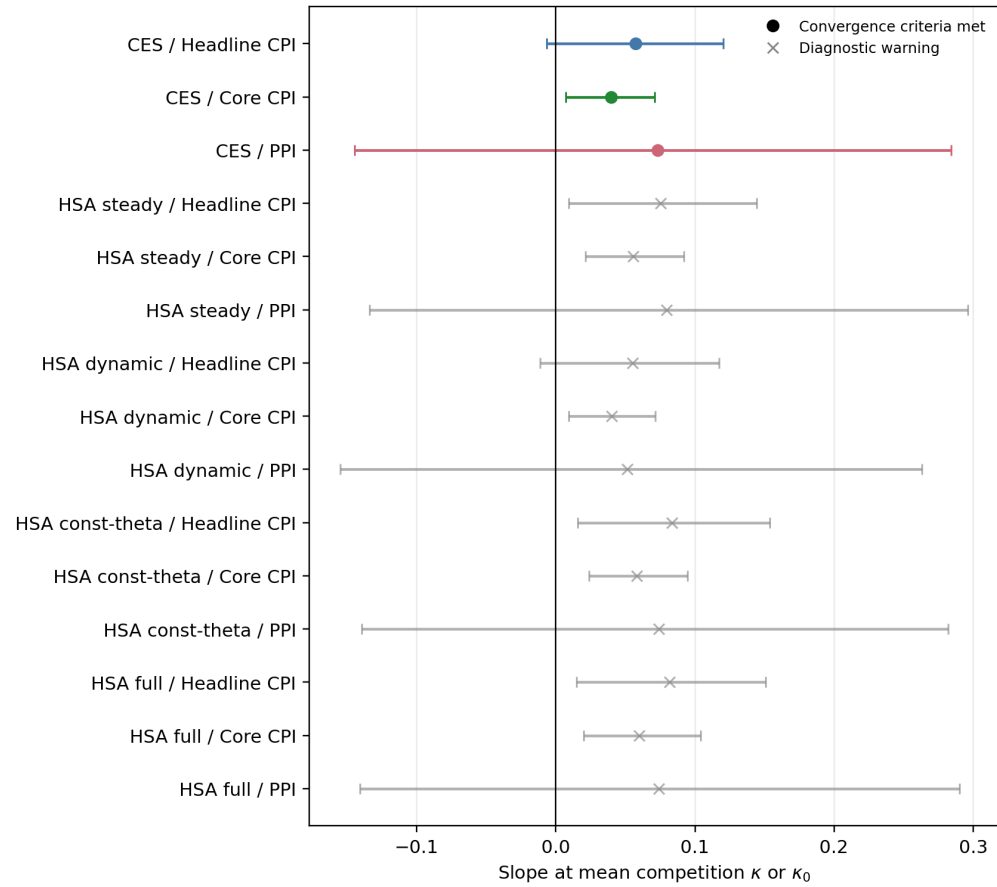


Figure 7: Annual-Q4 version: slope at average competition for the negative unemployment gap. Circles are runs meeting the convergence criteria; gray crosses are runs to watch.

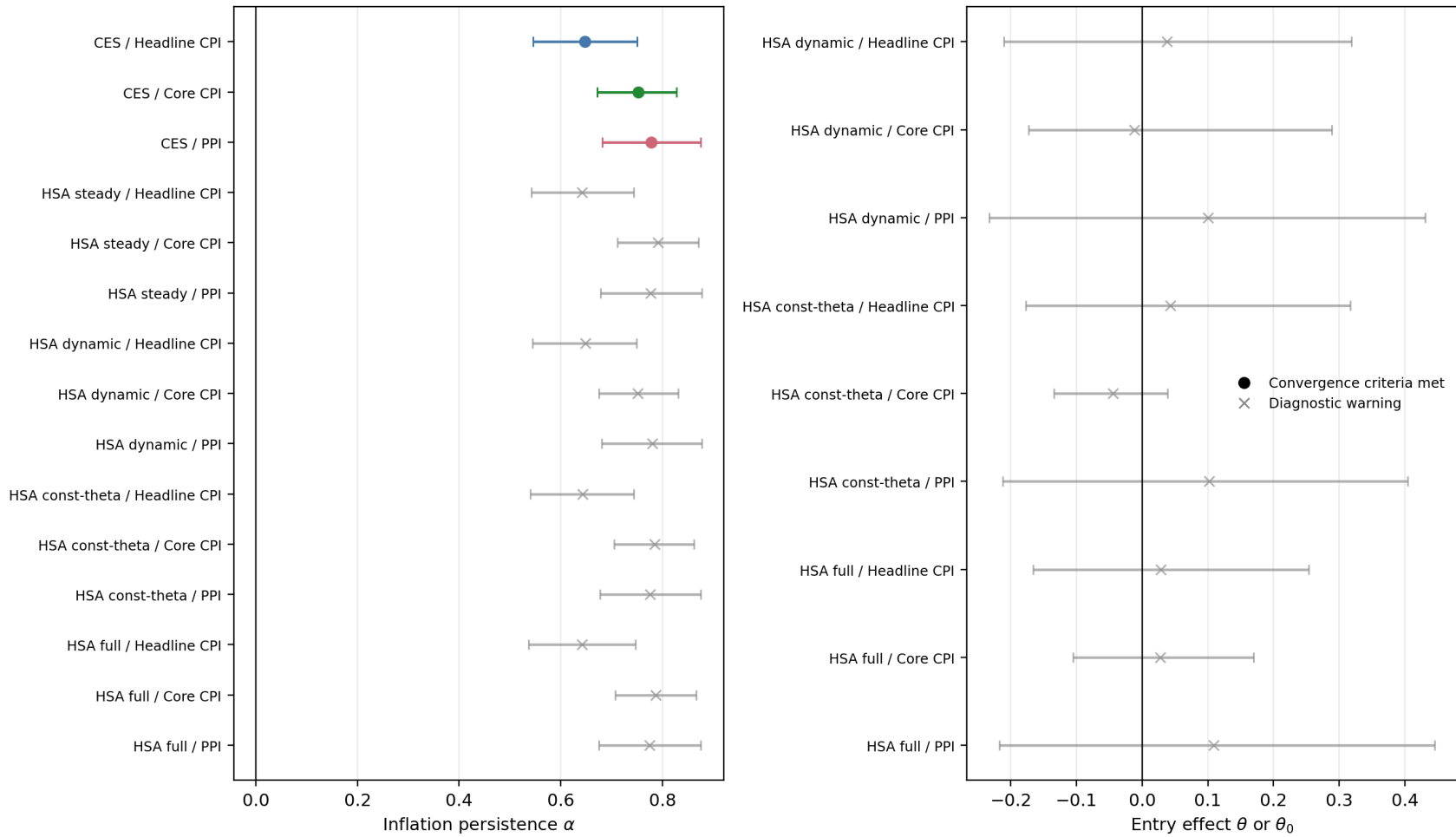


Figure 8: Annual-Q4 version: inflation inertia α (left) and entry coefficient θ or θ_0 (right). It shows whether the entry coefficient is updated by the data once the quarterly firm-count cycle is estimated inside the model.

7.4 CES–HSA slope bias

For the inverse-markup specification that corresponds directly to the theory,

$$\begin{aligned}\kappa_{\text{CES}} &= +0.007[-0.217, +0.228], & \kappa_{\text{HSA}} &= +0.010[-0.212, +0.223], \\ \kappa_{\text{CES}} - \kappa_{\text{HSA}} &= -0.003[-0.306, +0.311], & \Pr(\kappa_{\text{CES}} - \kappa_{\text{HSA}} < 0) &= 0.513, \\ B_{\text{HSA}} &= +0.001[-0.014, +0.015], & \Pr(B_{\text{HSA}} < 0) &= 0.424.\end{aligned}$$

The CES coefficient is identical to the PCHIP version, while the HSA coefficient and the $\hat{N}_t$ of (31) are computed from the annual-Q4 posterior. This keeps the smooth quarterly variation produced by interpolation out of the omitted-variable bias diagnostic. Because the annual-Q4 HSA dynamic, including the inverse markup, is outside the whole-run convergence criteria, the difference and OVB below are an additional diagnostic and not a formal test of the theoretical prediction.

Table 10: Annual-Q4 version: CES–HSA-dynamic slope difference and the omitted-variable bias implied by HSA.

specification	CES kappa	HSA kappa	CES-HSA	$\Pr(\text{CES-HSA}_t0)$	HSA-implied OVB	$\Pr(\text{OVB}_t0)$	diagnostics
Inverse markup (theory)	+0.007 [-0.217, +0.228] [†]	+0.010 [-0.212, +0.223] [†]	-0.003 [-0.306, +0.311] [†]	0.513	+0.001 [-0.014, +0.015] [†]	0.424	watch [†]
Headline CPI / negative unemployment gap	+0.057 [-0.006, +0.120] [†]	+0.055 [-0.011, +0.117] [†]	+0.002 [-0.089, +0.091] [†]	0.482	+0.002 [-0.003, +0.023] [†]	0.393	watch [†]
Core CPI / negative unemployment gap	+0.040 [+0.007, +0.071] [†]	+0.040 [+0.009, +0.071] [†]	-0.000 [-0.044, +0.045] [†]	0.511	-0.000 [-0.002, +0.002] [†]	0.687	watch [†]
PPI / negative unemployment gap	+0.073 [-0.145, +0.284] [†]	+0.051 [-0.155, +0.263] [†]	+0.022 [-0.279, +0.318] [†]	0.441	+0.002 [-0.005, +0.012] [†]	0.310	watch [†]

7.5 Prior sensitivity and convergence

The baseline/weak/tight comparison is also re-estimated under the same annual-Q4 observation scheme. This is not a re-weighting that appends a different prior to the PCHIP posterior: the firm-count states, coefficients, and variances are re-drawn by Gibbs/FFBS under each prior.

Table 11: Annual-Q4 version: all models, three-prior sensitivity for the negative unemployment gap. † marks failure of the convergence criteria.

model	inflation	parameter	baseline	weak	tight
CES	Core CPI	$kappa$	+0.040 [+0.007, +0.071]	+0.040 [+0.010, +0.071]	+0.039 [+0.002, +0.076]
CES	Headline CPI	$kappa$	+0.057 [-0.006, +0.120]	+0.056 [-0.011, +0.121]	+0.059 [-0.006, +0.124]
CES	PPI	$kappa$	+0.073 [-0.145, +0.284]	+0.060 [-0.181, +0.303]	+0.088 [-0.061, +0.241]
HSA const-theta	Core CPI	$delta$	+0.022 [+0.008, +0.038]†	+0.032 [+0.014, +0.065]†	+0.008 [-0.009, +0.023]†
HSA const-theta	Headline CPI	$delta$	+0.028 [-0.001, +0.055]†	+0.051 [+0.003, +0.103]†	+0.009 [-0.009, +0.027]†
HSA const-theta	PPI	$delta$	+0.011 [-0.026, +0.048]†	+0.072 [-0.058, +0.201]†	+0.002 [-0.017, +0.022]†
HSA dynamic	Core CPI	$kappa$	+0.040 [+0.009, +0.071]†	+0.033 [-0.007, +0.070]†	+0.040 [+0.005, +0.076]†
HSA dynamic	Headline CPI	$kappa$	+0.055 [-0.011, +0.117]†	+0.047 [-0.036, +0.130]†	+0.060 [-0.002, +0.123]
HSA dynamic	PPI	$kappa$	+0.051 [-0.155, +0.263]†	+0.020 [-0.249, +0.268]†	+0.083 [-0.072, +0.236]†
HSA full	Core CPI	$delta$	+0.024 [+0.009, +0.041]†	+0.028 [+0.009, +0.048]†	+0.007 [-0.009, +0.024]†
HSA full	Headline CPI	$delta$	+0.028 [+0.002, +0.054]†	+0.037 [-0.016, +0.084]†	+0.009 [-0.009, +0.028]†
HSA full	PPI	$delta$	+0.009 [-0.028, +0.047]†	+0.075 [-0.056, +0.200]†	+0.002 [-0.017, +0.022]†
HSA steady	Core CPI	$delta$	+0.022 [+0.008, +0.038]†	+0.038 [+0.013, +0.124]†	+0.009 [-0.007, +0.024]†
HSA steady	Headline CPI	$delta$	+0.029 [+0.005, +0.053]†	+0.052 [+0.017, +0.116]†	+0.010 [-0.008, +0.027]†
HSA steady	PPI	$delta$	+0.009 [-0.028, +0.047]†	+0.075 [-0.039, +0.188]†	+0.002 [-0.017, +0.021]†

Table 12: Model-by-model convergence tally for the annual-Q4 comparison. The 16 CES cells are a common reference; the 61 HSA cells are newly estimated.

model	runs	warnings	max Rhat	min bulk ESS
CES	16	0	1.002	2860
HSA const-theta	15	15	1.833	3
HSA dynamic	16	15	1.213	7
HSA full	15	15	1.942	3
HSA steady	15	15	1.749	3

For the annual-Q4 version too, the conclusions we adopt are limited to cells meeting the criteria of Table 12. In particular, a model that jointly identifies the quarterly cycle $\hat{N}_t$ and its coefficient from a firm count observed only annually carries little information. The widening intervals and lower ESS are themselves evidence of the measurement uncertainty hidden when PCHIP is taken as given. Also, because the δ interval for the CPI negative-unemployment-gap specification includes zero under the HSA-steady tight prior, the annual-Q4 version is not prior-invariant either. The same comparison by output gap is in Appendix B.

8 Scope condition and the limits of identification

For honesty we state where the main result does *not* hold and what it can and cannot claim.

- **Not identified for PPI.** Switching to PPI, closer to the theory’s left-hand side, makes δ include zero for all activity measures. Despite the measurement limitation that no PPI expectation exists, we cannot claim that the CPI relationship holds generally at the price-setting stage.
- **The complex models do not converge.** Many const-theta/full cells are outside the acceptance criteria, so we cannot claim to have demonstrated the cyclical-entry channel or γ .
- **The annual-Q4 joint posterior does not converge.** The CPI δ marginal chains mix well and give a positive point estimate close to the PCHIP version, but the ESS/Rhat of the AR(2) coefficients and latent-state paths are outside the criteria. The annual-Q4 result is therefore an important additional diagnostic of interpolation dependence, but at this stage does not count as a robustness check that confirms the main result.

Not a causal effect. $\bar{N}_t$ is a near-monotone low-frequency trend, so on this design δ cannot be separated from other things that trended over the same decades: the Volcker monetary-policy regime change, the anchoring of inflation expectations, changes in labor-market institutions and unionization, globalization and import competition, changes in the frequency of price adjustment, or a simple time trend / structural break. The most we can say is that *the low-frequency decline in the firm count and the decline in inflation’s sensitivity to the negative unemployment gap occurred together, with a sign consistent with the HSA theory*. We therefore avoid “competition causes flattening” and “we identify the mechanism.”

9 Robustness

We judge robustness on three points: (i) how the sign, interval, and magnitude change under alternative priors, (ii) whether the result survives switching the price index to PPI, and (iii) whether the same conclusion holds using only converged runs. The prior check re-estimates *every cell* under baseline, weak, and tight priors (Table 13); it is a genuine re-estimation of the current runs, not a re-weighting of a single posterior.

Table 13: Three-prior sensitivity of the current results (negative-unemployment-gap specification): δ under baseline, weak, and tight priors, by model and price index. Each cell is a separate re-estimation of the current runs, not a re-weighting. $\dagger$ marks cells outside the convergence criteria and not interpreted.

model	inflation	parameter	baseline	weak	tight
CES	Core CPI	$kappa$	+0.040 [+0.007, +0.071]	+0.040 [+0.010, +0.071]	+0.039 [+0.002, +0.076]
CES	Headline CPI	$kappa$	+0.057 [-0.006, +0.120]	+0.056 [-0.011, +0.121]	+0.059 [-0.006, +0.124]
CES	PPI	$kappa$	+0.073 [-0.145, +0.284]	+0.060 [-0.181, +0.303]	+0.088 [-0.061, +0.241]
HSA const-theta	Core CPI	$delta$	+0.031 [+0.012, +0.050] $\dagger$	+0.035 [+0.019, +0.052] $\dagger$	+0.008 [-0.008, +0.024] $\dagger$
HSA const-theta	Headline CPI	$delta$	+0.030 [-0.008, +0.062] $\dagger$	+0.061 [+0.027, +0.096] $\dagger$	+0.011 [-0.007, +0.029] $\dagger$
HSA const-theta	PPI	$delta$	+0.011 [-0.026, +0.048] $\dagger$	+0.092 [-0.036, +0.221] $\dagger$	+0.003 [-0.017, +0.022] $\dagger$
HSA dynamic	Core CPI	$kappa$	+0.032 [-0.004, +0.066]	+0.034 [-0.006, +0.071]	+0.031 [-0.009, +0.071]
HSA dynamic	Headline CPI	$kappa$	+0.052 [-0.026, +0.143] $\dagger$	+0.075 [-0.012, +0.161] $\dagger$	+0.043 [-0.027, +0.109]
HSA dynamic	PPI	$kappa$	+0.057 [-0.153, +0.276]	+0.045 [-0.247, +0.348] $\dagger$	+0.084 [-0.065, +0.245]
HSA full	Core CPI	$delta$	+0.025 [+0.007, +0.043] $\dagger$	+0.047 [+0.004, +0.134] $\dagger$	+0.016 [+0.002, +0.029] $\dagger$
HSA full	Headline CPI	$delta$	+0.033 [+0.006, +0.061] $\dagger$	+0.044 [-0.010, +0.119] $\dagger$	+0.014 [-0.003, +0.030] $\dagger$
HSA full	PPI	$delta$	+0.011 [-0.028, +0.049] $\dagger$	+0.073 [-0.063, +0.207] $\dagger$	+0.004 [-0.015, +0.023] $\dagger$
HSA steady	Core CPI	$delta$	+0.023 [+0.010, +0.039]	+0.037 [+0.013, +0.109]	+0.013 [+0.003, +0.024]
HSA steady	Headline CPI	$delta$	+0.030 [+0.008, +0.051]	+0.056 [+0.018, +0.133]	+0.016 [+0.000, +0.030]
HSA steady	PPI	$delta$	+0.012 [-0.024, +0.049]	+0.073 [-0.061, +0.189]	+0.004 [-0.015, +0.024]

HSA steady. The headline-CPI and core-CPI δ keep a positive sign under baseline/weak/tight, with posterior intervals on the positive side. But the interval is wide under weak and shrinks toward zero under tight, so it is not fully prior-invariant. The core-CPI baseline carries the strongest weight of evidence. PPI includes zero under all three priors and does not provide external confirmation of the CPI result.

Model variants. The $\dagger$ cells of const-theta/full are not counted as robustness evidence even when the sign matches steady. Dynamic is also short of ESS in some cells. Before claiming “the result survives” under added complexity, convergence must be established through reparameterization or longer independent chains.

Period robustness. The state equation assumes consecutive quarters. We do not use the old treatment that deletes an interior span and compresses what precedes and follows. The current 77 runs are full-sample comparisons; until a separate contiguous subsample estimation of the corrected revision is done, we do not include pre-2008 or post-1988 numbers in the conclusions.

10 Conclusion

Over 1982–2012 in the United States, the low-frequency decline in the number of listed firms and the decline in inflation’s sensitivity to the negative unemployment gap moved together. We read this as theory-consistent time-series evidence, not a causal effect.

- **Sign consistent with HSA.** For the core-CPI negative-unemployment-gap specification $\delta = +0.023[+0.010, +0.039]$ ($\text{BF}_{10} = 429.7$); a falling firm count implies a flatter slope, as the theory predicts.
- **Economically sizable flattening.** The implied κ_t falls from $+0.120$ to -0.005 ; the start-to-end response to a one-point negative unemployment gap differs by about 0.126 point.
- **Core CPI aligns the output gap.** Under core CPI the HP output gap shows the same positive sign; BN stays inconclusive.
- **Not confirmed for PPI.** The negative-unemployment-gap specification gives $\delta = +0.012[-0.024, +0.049]$, with the interval including zero for all activity measures, so the relationship is not a general upstream (price-setting-stage) effect.
- **No CES-slope bias found.** The theory’s predicted negative bias of the CES slope is not supported: both the CES–HSA difference and the HSA-implied omitted-variable bias have intervals including zero, with, if anything, positive point estimates.
- **Cyclical-entry channel unconfirmed.** Many HSA const-theta/full cells do not converge and the converged entry coefficients include zero.
- **Annual-Q4 is provisional.** The CPI δ marginal posterior stays positive under Q4-only observation, but the latent states and AR(2) block miss the convergence criteria, so this is a sensitivity diagnostic, not a confirmed robustness check.
- **Identification caveat.** The variation is concentrated in the 1980s and $\bar{N}_t$ is nearly collinear with a linear trend, so δ is an association, not a causal effect.

11 Status and open items

Estimation status (auto-generated). This draft uses the 77 runs of the current corrected estimation revision. Of these, 29 runs fail the acceptance criteria (maximum $\hat{R} \leq 1.01$ and minimum bulk ESS ≥ 400) and their coefficients are not used for the main conclusions. All headline estimates, tables, and figures are generated automatically from the saved posteriors; no numbers from earlier revisions are read in. In addition, an annual-Q4 variant—in which the firm count is observed only in Q4 and the intervening quarterly states are inferred—is assembled from *new* estimates of the 61 HSA cells and a shared reference of 16 CES cells that do not use the firm count. Of the 77 cells in the annual-Q4 comparison, 60 fail the same convergence criteria.

Settled in the current revision. The positive δ and flattening κ_t for the CPI negative-unemployment-gap specification of HSA steady; the positive sign of the core-CPI HP specification and the non-result of BN, separated into the appendix; the non-results of the three PPI activity specifications; the three-prior sensitivity; the convergence of CES/HSA steady; the economic magnitude; and the CES–HSA slope-bias diagnostic via the inverse markup. The HSA results settled here are the PCHIP version. The 61 annual-Q4 HSA cells are estimated in this setting but retain the convergence issues below.

Open (empirical / econometric).

- **Causal identification.** The horse race against a linear time trend cannot distinguish the two. Before any claim beyond association, add quadratic trends, structural-break dummies, and placebo interactions.
- **Expectation mismatch.** Beyond core-matched expected inflation, connect a PPI forecast series (e.g. the Livingston Survey) so the expectation term matches the price index.
- **Predictive evaluation.** Replace the in-sample plug-in fit score with LOO/WAIC or rolling out-of-sample forecasts.
- **Period robustness.** Re-estimate the current revision using only contiguous quarters. The pre-2008/post-1988 numbers from earlier revisions are not used for the conclusions of this paper.
- **Bayes-factor robustness.** In addition to the two-sided SDDR, report the one-sided ($\delta > 0$) evaluation and the KDE bandwidth sensitivity.
- **Annual-Q4 state block.** The marginal chains of δ, κ_0 are good, but ρ_1, ρ_2 and some time points of $\bar{N}_t, \hat{N}_t$ have low ESS / high Rhat. Consider reparameterization, longer independent chains, and priors that relax the weak identification of the state equation.

Open (computational).

- **Automatic marginal likelihood.** The Chib marginal likelihood is excluded from the automatic tables because the integration over the latent states and the normalization of the truncated prior are unverified. It is not used for model selection until a verified implementation exists.

- **Weak convergence of const-theta/full.** The cells to watch are dropped from the main inference. Consider reparameterization, longer independent chains, and identifying restrictions.
- **Weak identification of γ .** An identification strategy is needed to relax the collinearity between $\hat{N}\bar{N}$ and $\hat{N}$.

A Appendix: results by output gap

The body takes the negative unemployment gap as the main activity measure and separates the output-gap results here. Because HP and BN estimate potential output differently, we do not combine them into a single “output gap” coefficient but report the full baseline results by price index (3) and model (5) for each definition. The slope is the constant κ for CES/HSA dynamic and the slope at average competition κ_0 for the other HSA models. δ is the sensitivity of the slope to the firm-count trend, and entry denotes θ or θ_0 .

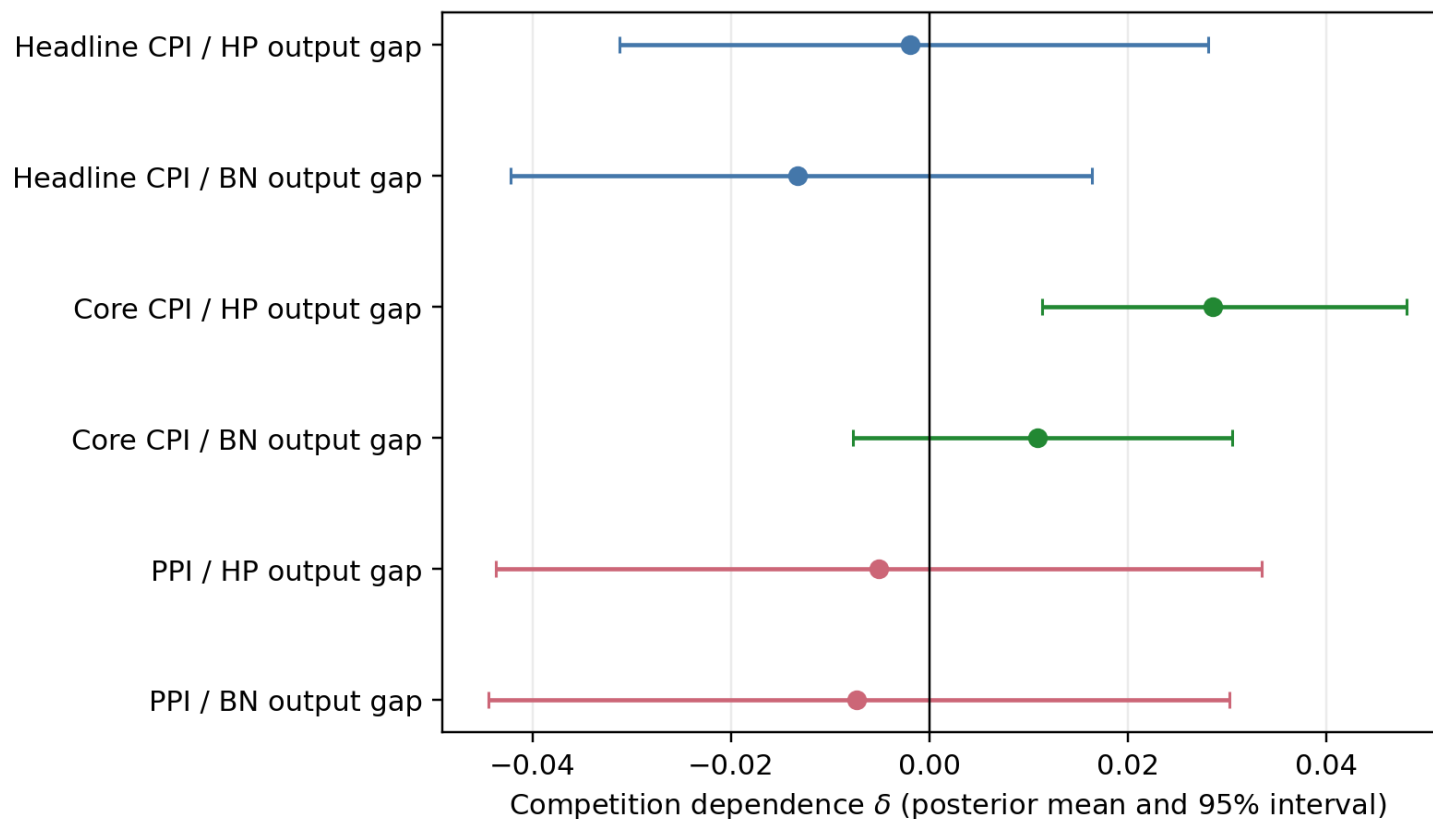


Figure 9: HSA-steady δ by output gap (posterior mean and 95% CI). HP and BN are separated and headline CPI, core CPI, and PPI are shown on the same scale.

A.1 HP output gap

For HSA steady the headline-CPI δ is near zero, while core CPI is positive:

$$\delta_{\text{Core,HP}} = +0.029[+0.011, +0.048], \quad \text{BF}_{10} = 37.3.$$

The PPI $\delta = -0.005[-0.044, +0.034]$ includes zero. The positive competition dependence of the HP specification is therefore limited to core CPI and is not a result for price indices in general.

Table 14: HP output gap: baseline estimates by price index and model. † marks failure of the convergence criteria.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.140 [+0.050, +0.232]	—	—	—	—	—	OK
CES	Core CPI	+0.091 [+0.042, +0.140]	—	—	—	—	—	OK
CES	PPI	+0.210 [-0.062, +0.470]	—	—	—	—	—	OK
HSA steady	Headline CPI	+0.141 [+0.048, +0.237]	-0.002 [-0.031, +0.028]	0.8	—	—	+0.135 → +0.146	OK
HSA steady	Core CPI	+0.069 [+0.018, +0.123]	+0.029 [+0.011, +0.048]	37.3	—	—	+0.161 → -0.005	OK
HSA steady	PPI	+0.210 [-0.067, +0.473]	-0.005 [-0.044, +0.034]	1.0	—	—	+0.193 → +0.224	OK
HSA dynamic	Headline CPI	+0.130 [+0.038, +0.222]	—	—	-0.026 [-0.123, +0.106]	—	—	OK
HSA dynamic	Core CPI	+0.085 [+0.037, +0.132]	—	—	-0.025 [-0.072, +0.029]	—	—	OK
HSA dynamic	PPI	+0.163 [-0.114, +0.437]	—	—	+0.019 [-0.268, +0.328]	—	—	OK
HSA const-theta	Headline CPI	+0.145 [+0.050, +0.239]†	-0.003 [-0.037, +0.034]†	0.9	+0.032 [-0.112, +0.159]†	—	+0.139 → +0.151	watch†
HSA const-theta	Core CPI	+0.079 [+0.031, +0.127]†	+0.033 [+0.012, +0.055]†	56.5	+0.005 [-0.048, +0.060]†	—	+0.158 → +0.003	watch†
HSA const-theta	PPI	+0.219 [-0.049, +0.494]	-0.004 [-0.043, +0.035]	1.1	+0.056 [-0.221, +0.338]	—	+0.209 → +0.228	OK
HSA full	Headline CPI	+0.136 [+0.040, +0.233]†	-0.006 [-0.037, +0.028]†	0.9	+0.029 [-0.100, +0.147]†	-0.019 [-0.057, +0.022]†	+0.115 → +0.151	watch†
HSA full	Core CPI	+0.083 [+0.022, +0.192]†	+0.028 [+0.006, +0.052]†	13.0	+0.022 [-0.046, +0.092]†	-0.014 [-0.045, +0.017]†	+0.145 → +0.006	watch†
HSA full	PPI	+0.213 [-0.059, +0.488]†	-0.004 [-0.044, +0.033]†	1.0	+0.018 [-0.271, +0.318]†	-0.005 [-0.044, +0.034]†	+0.199 → +0.224	watch†

A.2 BN output gap

For HSA steady the δ point estimates are negative for headline CPI, positive for core CPI, and negative for PPI, but all three 95% credible intervals include zero (headline $-0.013[-0.042, +0.016]$, core $+0.011[-0.008, +0.031]$, PPI $-0.007[-0.044, +0.030]$). Each BF_{10} is also about one, so the BN specification provides no evidence for a competition dependence of the slope.

Table 15: BN output gap: baseline estimates by price index and model. † marks failure of the convergence criteria.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.059 [-0.025, +0.144]	—	—	—	—	—	OK
CES	Core CPI	+0.032 [-0.018, +0.082]	—	—	—	—	—	OK
CES	PPI	+0.189 [-0.049, +0.428]	—	—	—	—	—	OK
HSA steady	Headline CPI	+0.064 [-0.024, +0.153]	-0.013 [-0.042, +0.016]	1.1	—	—	+0.022 → +0.097	OK
HSA steady	Core CPI	+0.031 [-0.020, +0.083]	+0.011 [-0.008, +0.031]	1.0	—	—	+0.063 → +0.005	OK
HSA steady	PPI	+0.190 [-0.054, +0.427]	-0.007 [-0.044, +0.030]	1.1	—	—	+0.167 → +0.210	OK
HSA dynamic	Headline CPI	+0.057 [-0.026, +0.137]	—	—	-0.076 [-0.170, +0.083]	—	—	OK
HSA dynamic	Core CPI	+0.034 [-0.012, +0.081]	—	—	-0.040 [-0.087, +0.013]	—	—	OK
HSA dynamic	PPI	+0.159 [-0.098, +0.404]	—	—	-0.040 [-0.325, +0.300]	—	—	OK
HSA const-theta	Headline CPI	+0.059 [-0.030, +0.145]†	-0.013 [-0.044, +0.019]†	1.1	-0.008 [-0.126, +0.114]†	—	+0.033 → +0.086	watch†
HSA const-theta	Core CPI	+0.028 [-0.022, +0.079]†	+0.015 [-0.006, +0.036]†	1.4	-0.043 [-0.099, +0.013]†	—	+0.066 → -0.003	watch†
HSA const-theta	PPI	+0.188 [-0.049, +0.430]†	-0.008 [-0.046, +0.029]†	1.0	-0.006 [-0.297, +0.285]†	—	+0.167 → +0.209	watch†
HSA full	Headline CPI	+0.071 [-0.041, +0.168]†	-0.008 [-0.042, +0.036]†	1.2	+0.051 [-0.092, +0.170]†	-0.016 [-0.062, +0.034]†	+0.044 → +0.103	watch†
HSA full	Core CPI	+0.046 [-0.004, +0.096]†	+0.012 [-0.005, +0.030]†	1.1	+0.024 [-0.061, +0.110]†	-0.033 [-0.064, -0.003]†	+0.086 → +0.014	watch†
HSA full	PPI	+0.198 [-0.039, +0.444]†	-0.006 [-0.045, +0.033]†	1.1	+0.052 [-0.261, +0.378]†	-0.005 [-0.043, +0.034]†	+0.184 → +0.212	watch†

How to read the whole. HP core CPI agrees in direction with the CPI result for the negative unemployment gap, but HP headline, HP PPI, and all price indices of BN do not support it. And because const-theta/full have many cells outside the convergence criteria, we do not count sign agreement in the complex models as robustness evidence. The output-gap results are not a claim that “the result always holds across activity measures” but a sensitivity analysis showing which combinations of measurement and price index the result depends on.

B Appendix: annual-Q4 results by output gap

We apply the same mixed-frequency estimation as Section 7 to the HP and BN output gaps. Each table gives the full baseline results by price index (3) and model (5); only the CES row is a common reference that does not depend on the firm-count observation scheme. We first compare the HSA-steady δ on the same scale, then show the full coefficients and diagnostics by model.

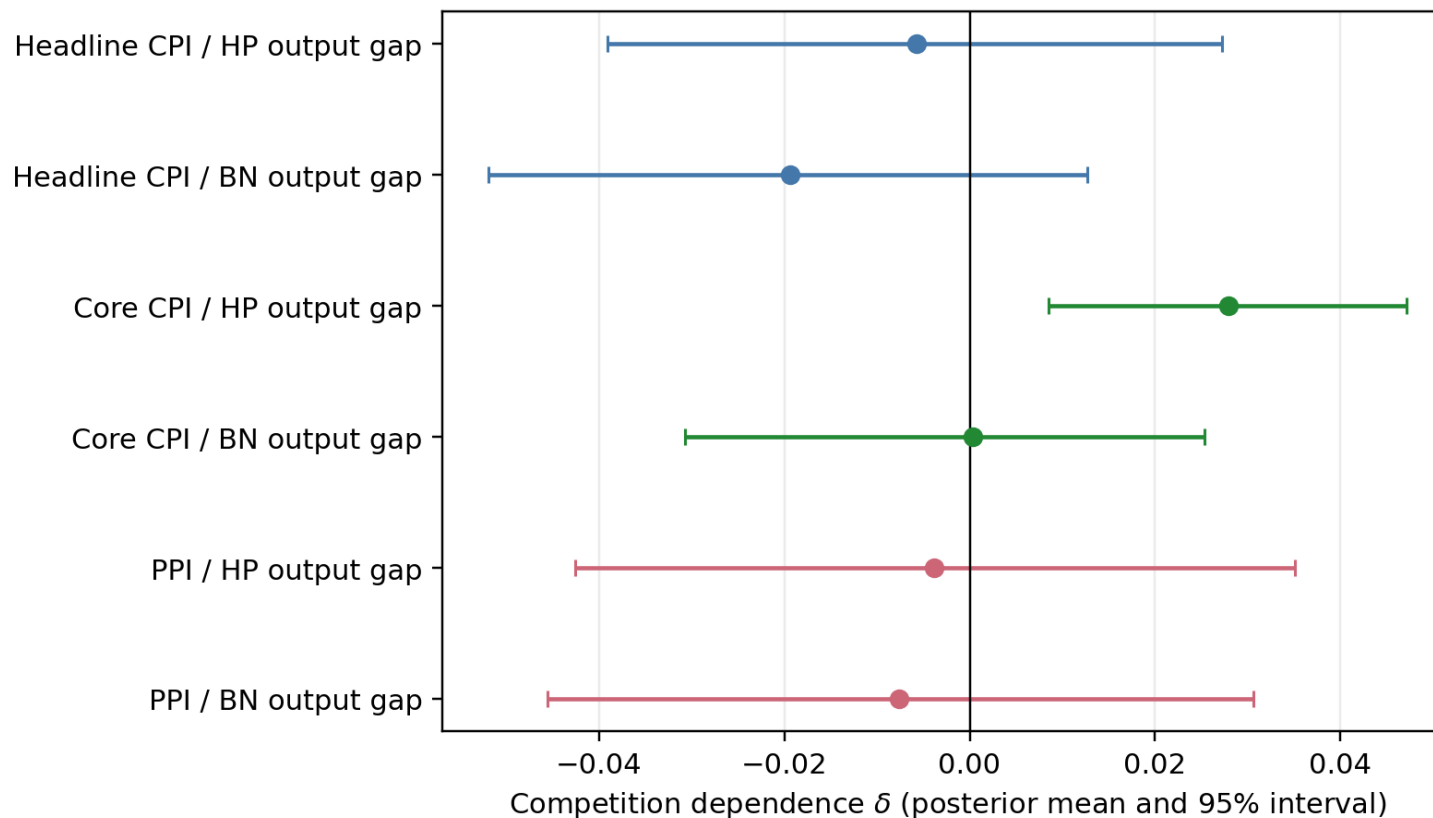


Figure 10: Annual-Q4 version: HSA-steady δ by HP and BN output gap (posterior mean and 95% CI).

B.1 HP output gap

For the annual-Q4 HSA steady, headline CPI gives $\delta = -0.006[-0.039, +0.027]$ ($\text{BF}_{10} = 1.0$), core CPI gives $\delta = +0.028[+0.009, +0.047]$ ($\text{BF}_{10} = 19.5$), and PPI gives $\delta = -0.004[-0.043, +0.035]$ ($\text{BF}_{10} = 1.0$).

Table 16: Annual-Q4 version: HP output gap, baseline estimates by price index and model.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.140 [+0.050, +0.232]	–	–	–	–	–	OK
CES	Core CPI	+0.091 [+0.042, +0.140]	–	–	–	–	–	OK
CES	PPI	+0.210 [–0.062, +0.470]	–	–	–	–	–	OK
HSA steady	Headline CPI	+0.140 [+0.048, +0.231] [†]	–0.006 [–0.039, +0.027] [†]	1.0	–	–	+0.135 → +0.155	watch [†]
HSA steady	Core CPI	+0.073 [+0.022, +0.124] [†]	+0.028 [+0.009, +0.047] [†]	19.5	–	–	+0.152 → +0.001	watch [†]
HSA steady	PPI	+0.212 [–0.040, +0.469] [†]	–0.004 [–0.043, +0.035] [†]	1.0	–	–	+0.207 → +0.221	watch [†]
HSA dynamic	Headline CPI	+0.139 [+0.052, +0.226] [†]	–	–	+0.054 [–0.169, +0.346] [†]	–	–	watch [†]
HSA dynamic	Core CPI	+0.090 [+0.046, +0.135] [†]	–	–	+0.014 [–0.183, +0.320] [†]	–	–	watch [†]
HSA dynamic	PPI	+0.162 [–0.109, +0.445] [†]	–	–	+0.112 [–0.215, +0.454] [†]	–	–	watch [†]
HSA const-theta	Headline CPI	+0.141 [+0.049, +0.232] [†]	–0.006 [–0.038, +0.027] [†]	0.9	+0.043 [–0.167, +0.304] [†]	–	+0.135 → +0.156	watch [†]
HSA const-theta	Core CPI	+0.081 [+0.031, +0.132] [†]	+0.023 [–0.013, +0.048] [†]	3.0	–0.010 [–0.137, +0.235] [†]	–	+0.131 → +0.024	watch [†]
HSA const-theta	PPI	+0.212 [–0.061, +0.478] [†]	–0.004 [–0.043, +0.035] [†]	1.0	+0.109 [–0.236, +0.471] [†]	–	+0.207 → +0.222	watch [†]
HSA full	Headline CPI	+0.139 [+0.051, +0.226] [†]	–0.007 [–0.041, +0.027] [†]	0.9	+0.067 [–0.214, +0.404] [†]	–0.000 [–0.041, +0.039] [†]	+0.138 → +0.156	watch [†]
HSA full	Core CPI	+0.072 [+0.019, +0.123] [†]	+0.028 [+0.009, +0.048] [†]	33.9	–0.009 [–0.109, +0.087] [†]	–0.007 [–0.039, +0.028] [†]	+0.154 → –0.002	watch [†]
HSA full	PPI	+0.209 [–0.062, +0.479] [†]	–0.004 [–0.043, +0.036] [†]	1.1	+0.106 [–0.258, +0.473] [†]	–0.000 [–0.038, +0.037] [†]	+0.206 → +0.218	watch [†]

B.2 BN output gap

For the annual-Q4 HSA steady, headline CPI gives $\delta = -0.019[-0.052, +0.013]$ ($\text{BF}_{10} = 1.6$), core CPI gives $\delta = +0.000[-0.031, +0.025]$ ($\text{BF}_{10} = 0.8$), and PPI gives $\delta = -0.008[-0.046, +0.031]$ ($\text{BF}_{10} = 1.0$).

Table 17: Annual-Q4 version: BN output gap, baseline estimates by price index and model.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.059 [–0.025, +0.144]	–	–	–	–	–	OK
CES	Core CPI	+0.032 [–0.018, +0.082]	–	–	–	–	–	OK
CES	PPI	+0.189 [–0.049, +0.428]	–	–	–	–	–	OK
HSA steady	Headline CPI	+0.053 [–0.033, +0.139] [†]	–0.019 [–0.052, +0.013] [†]	1.6	–	–	+0.037 → +0.103	watch [†]
HSA steady	Core CPI	+0.030 [–0.021, +0.081] [†]	+0.000 [–0.031, +0.025] [†]	0.8	–	–	+0.041 → +0.030	watch [†]
HSA steady	PPI	+0.187 [–0.051, +0.437] [†]	–0.008 [–0.046, +0.031] [†]	1.0	–	–	+0.177 → +0.207	watch [†]
HSA dynamic	Headline CPI	+0.059 [–0.025, +0.142] [†]	–	–	+0.055 [–0.158, +0.323] [†]	–	–	watch [†]
HSA dynamic	Core CPI	+0.030 [–0.015, +0.077] [†]	–	–	–0.019 [–0.163, +0.262] [†]	–	–	watch [†]
HSA dynamic	PPI	+0.156 [–0.091, +0.400] [†]	–	–	+0.117 [–0.227, +0.466] [†]	–	–	watch [†]
HSA const-theta	Headline CPI	+0.051 [–0.040, +0.136] [†]	–0.019 [–0.051, +0.014] [†]	1.5	+0.056 [–0.174, +0.352] [†]	–	+0.034 → +0.101	watch [†]
HSA const-theta	Core CPI	+0.029 [–0.023, +0.080] [†]	+0.001 [–0.030, +0.026] [†]	0.7	–0.015 [–0.145, +0.251] [†]	–	+0.040 → +0.026	watch [†]
HSA const-theta	PPI	+0.185 [–0.047, +0.421] [†]	–0.007 [–0.046, +0.031] [†]	1.1	+0.110 [–0.252, +0.462] [†]	–	+0.180 → +0.202	watch [†]
HSA full	Headline CPI	+0.051 [–0.035, +0.135] [†]	–0.020 [–0.052, +0.014] [†]	1.6	+0.064 [–0.199, +0.375] [†]	–0.000 [–0.039, +0.038] [†]	+0.040 → +0.102	watch [†]
HSA full	Core CPI	+0.026 [–0.026, +0.078] [†]	–0.006 [–0.037, +0.020] [†]	0.8	+0.019 [–0.181, +0.354] [†]	–0.003 [–0.041, +0.038] [†]	+0.033 → +0.041	watch [†]
HSA full	PPI	+0.181 [–0.068, +0.426] [†]	–0.007 [–0.045, +0.032] [†]	1.1	+0.114 [–0.223, +0.450] [†]	+0.000 [–0.039, +0.040] [†]	+0.173 → +0.200	watch [†]

These correspond one-to-one with Appendix A of the PCHIP version. In the comparison we read not only the point estimate but the 95% credible interval, the Bayes factor, and the convergence flag together. Where an interval widens under annual-Q4, that is the result of not treating the quarterly firm count as observed and propagating the interpolation uncertainty.

C Appendix: full model-by-model tables (negative unemployment gap)

This appendix collects the dense, auto-generated tables that support Sections 5 and 9: the full model-by-model coefficients for the negative unemployment gap and the whole-run convergence tally. (The three-prior sensitivity table now sits with the robustness discussion, Table 13.) The slope is κ for CES/HSA dynamic and κ_0 for the other models; entry is θ or θ_0 ; † marks cells outside the convergence criteria.

Table 18: Model-by-model results for the negative-unemployment-gap specification. Slope is κ for CES/HSA dynamic and κ_0 otherwise; entry is θ or θ_0 . † marks cells that fail the convergence criteria and are not interpreted.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.057 [-0.006, +0.120]	—	—	—	—	—	OK
CES	Core CPI	+0.040 [+0.007, +0.071]	—	—	—	—	—	OK
CES	PPI	+0.073 [-0.145, +0.284]	—	—	—	—	—	OK
HSA steady	Headline CPI	+0.067 [+0.002, +0.136]	+0.030 [+0.008, +0.051]	22.8	—	—	+0.163 → -0.010	OK
HSA steady	Core CPI	+0.051 [+0.016, +0.090]	+0.023 [+0.010, +0.039]	429.7	—	—	+0.120 → -0.005	OK
HSA steady	PPI	+0.075 [-0.131, +0.279]	+0.012 [-0.024, +0.049]	1.2	—	—	+0.113 → +0.044	OK
HSA dynamic	Headline CPI	+0.052 [-0.026, +0.143]	—	—	-0.013 [-0.144, +0.166]	—	—	watch†
HSA dynamic	Core CPI	+0.032 [-0.004, +0.066]	—	—	-0.022 [-0.076, +0.039]	—	—	OK
HSA dynamic	PPI	+0.057 [-0.153, +0.276]	—	—	+0.011 [-0.296, +0.343]	—	—	OK
HSA const-theta	Headline CPI	+0.089 [+0.017, +0.163]	+0.030 [-0.008, +0.062]	4.1	+0.093 [-0.026, +0.208]	—	+0.146 → +0.049	watch†
HSA const-theta	Core CPI	+0.063 [+0.024, +0.101]	+0.031 [+0.012, +0.050]	49.8	+0.023 [-0.036, +0.089]	—	+0.123 → +0.001	watch†
HSA const-theta	PPI	+0.082 [-0.142, +0.304]	+0.011 [-0.026, +0.048]	1.1	+0.043 [-0.252, +0.331]	—	+0.111 → +0.055	watch†
HSA full	Headline CPI	+0.088 [+0.011, +0.161]	+0.033 [+0.006, +0.061]	11.6	+0.070 [-0.057, +0.195]	-0.015 [-0.052, +0.023]	+0.160 → +0.018	watch†
HSA full	Core CPI	+0.062 [+0.020, +0.105]	+0.025 [+0.007, +0.043]	19.9	+0.052 [-0.027, +0.132]	-0.010 [-0.043, +0.021]	+0.137 → -0.005	watch†
HSA full	PPI	+0.080 [-0.145, +0.301]	+0.011 [-0.028, +0.049]	1.1	+0.043 [-0.262, +0.373]	-0.005 [-0.043, +0.035]	+0.111 → +0.054	watch†

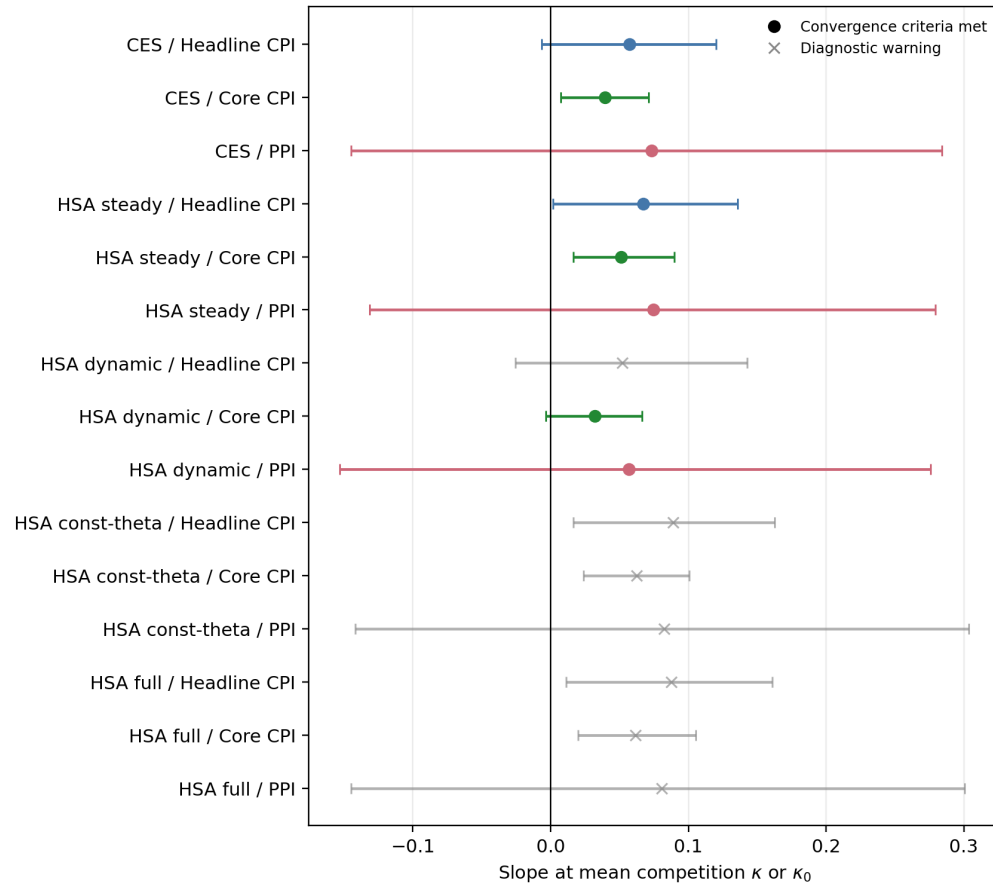


Figure 11: Slope at average competition for the negative-unemployment-gap specification. κ for CES/HSA dynamic and κ_0 otherwise. Circles are runs meeting the convergence criteria; gray crosses are runs to watch.

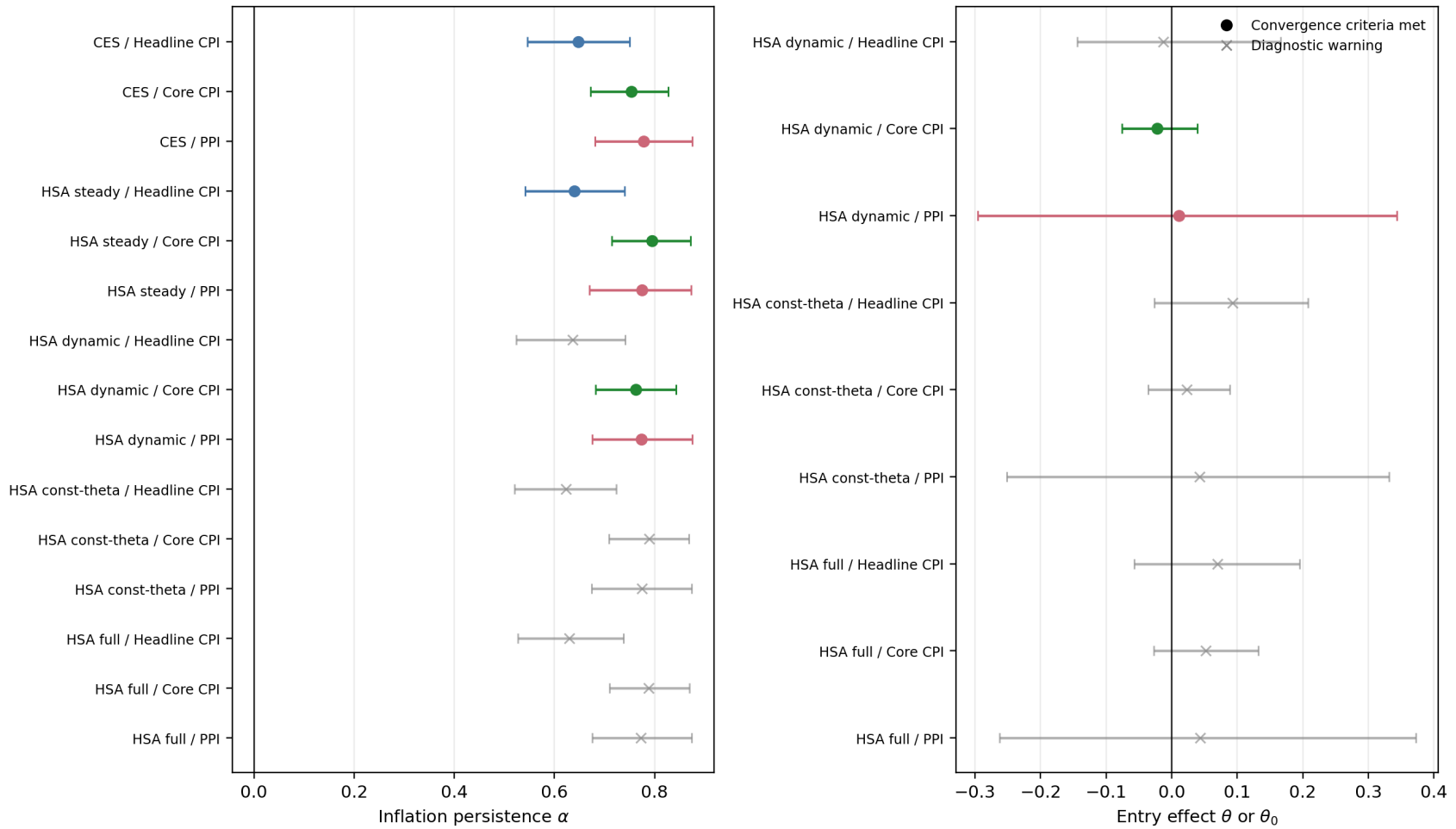


Figure 12: Inflation inertia α (left) and entry coefficient θ or θ_0 (right) for the negative-unemployment-gap specification. Circles and gray crosses are as in Figure 11.

Table 19: Model-by-model convergence tally across all 77 runs. “warnings” is the number of runs outside the acceptance criteria.

model	runs	warnings	max Rhat	min bulk ESS
CES	16	0	1.002	2860
HSA const-theta	15	14	1.840	3
HSA dynamic	16	3	1.009	146
HSA full	15	15	1.914	3
HSA steady	15	0	1.007	401